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Deferred Tax Asset Revaluations, Costly Information Processing, and Bank Deposits: Evidence from the Tax Cuts and Jobs Act

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Abstract. We examine how information processing costs affect the extent to which depositors' use the details in banks' income statements. Depositors have a unique cost-benefit structure because they are nonprofessional users of financial information and have high information processing costs. At the same time, they benefit from acting quickly because failing banks make payments on a first-come, first-serve basis. This makes it likely that they will react to a prominent summary measure like the reported net income without adjusting for any risk-irrelevant information included in the line items. In our empirical analysis, we investigate depositors' behavior as driven by the mechanical revaluations of deferred tax assets due to the 2017 Tax Cuts and Jobs Act. Using a difference-in-difference design, we find deposit withdrawals because of this risk-irrelevant information. In cross-sectional tests, we show that the withdrawals are stronger if the information acquisition costs are low, and the information integration costs are high. Overall, our results show that information processing costs are important for understanding depositors' reactions to accounting information and can lead to deposit flows that cannot be explained by new risk-relevant information.

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Keywords: financial accounting in banks • information processing costs • retail depositors • tax accruals

1. Introduction

In this paper, we explore whether and to what extent bank depositors use the details from bank income statements to inform their funding decisions. Although there is evidence that they use accounting information in their decision making (Chen et al. 2022), we argue, based on the rational inattention models reviewed in Blankespoor et al. (2020), that they do not fully process all available information in the income statement. Understanding the decision making of depositors is important because depositors are at the heart of the microeconomic theory of banks (Beatty and Liao 2014) and provide USD 19 trillion or 90% of banks' liabilities (FDIC 2022). This impact makes them an important stakeholder group who can influence banks' solvency and

liquidity (Flannery 2001, Morris and Shin 2002, Brown et al. 2017, Liang and Zhang 2019).

Information processing costs play an important role in understanding depositors' decision making. Theoretical models highlight crucial differences between information unaware depositors and depositors that are aware but poorly understand the information: On the one hand, depositors that are unaware of information will react less to information cascades than those who are aware (Chari and Jagannathan 1988, Calomiris and Khan 1991). On the other hand, if depositors are aware of the information but do not sufficiently invest time and effort into integrating the information, then they will react heuristically to a shared public signal.

The information levels in these models mirror the three different constraints to information processing that are related to awareness, acquisition, and integration as defined in Blankespoor et al. (2019, 2020). However, it is not possible to directly transfer the results on retail shareholders reviewed in Blankespoor et al. (2020) to the depositor setting because the costs and benefits of the processing of financial statement details differ in important ways. For example, the benefits of information gathering are lower because the deposit insurance reduces the downside risk, whereas the nature of the debt contract restricts the upside potential. At the same time, depositors can benefit from withdrawing deposits early without processing all information (Diamond and Dybvig 1983, Calomiris and Khan 1991, Chen 1999, Goldstein and Pauzner 2005). In this paper, we investigate how the combination of these factors shape the decision-making of bank depositors.

We use a difference-in-differences approach to compare the level of bank deposits before and after the 2017 Tax Cuts and Jobs Act (TCJA) (the first difference) and to distinguish between banks with different amounts of revaluations of deferred tax assets (DTAs) due to a change in the underlying tax rate due to the TCJA (the second difference). The DTA revaluations arising due to the TCJA offer a unique setting to isolate the information processing of accounting signals by depositors from their reactions to underlying economic fundamentals. The TCJA cut the maximum corporate income tax rate from 35% to 21% for all firms (Gaertner et al. 2020) and thus forced firms with DTAs to revalue these tax accruals that directly reduced bank profits in the income statement. Appendix A shows one real example of how banks displayed the information in the income statement. The TCJA affects banks with DTAs in two ways: On the one hand, the reduction in the tax rate increases future after-tax income. On the other hand, it reduces the deductions that the companies can claim with respect to past losses recorded in the form of DTAs. As long as the expected future profits are higher than the current loan loss provisions, the positive effect will dominate. Importantly, the effect on the net income can be inferred from the prior quarter's financial statements, so stakeholders can react immediately after the TCJA's announcement and the actual reporting of the revaluations does not contain new information. Moreover, despite their effects on the bottom-line net income, the DTA revaluations contain no forward-looking information on bank risk-taking (Laux 2013) and they do not change the underlying tax base (e.g., the net operating income of the current and future periods).¹ Consequently, if depositors sufficiently integrate the information from these DTA revaluations, we should observe no withdrawal of deposits at the time of the next earnings announcement.² However, the accounting rules recognize the losses upfront and

the benefits when they are realized. It requires effort to acquire and sufficiently integrate this financial information. Because of the high costs and the low benefits of information processing, we expect that depositors remain rationally inattentive and do not understand the risk-irrelevance of the change in tax accruals. Instead, they react heuristically to the negative net income by withdrawing deposits, in line with their incentive to act quickly. Because depositors have to adjust the net income to correctly interpret the accounting signal, we expect that information awareness and information integration costs work differently: Withdrawal of deposits due to the misinterpretation of the earnings signal is most likely if the information awareness costs are low but the information integration costs are high.

We use the quarterly call reports of all commercial banks without international operations³ from the first quarter of 2015 through the fourth quarter of 2019 to perform a difference-in-differences analysis with a proportional, nonbinary treatment that resembles a fuzzy difference-in-differences approach (Chaisemartin and D'Haultfœuille 2018). A difference-in-differences approach is helpful in our context because it alleviates concerns that other confounding influences from a simultaneous event might potentially contribute to the observed effects. The standard difference-in-differences approach views all treated firms the same. In contrast, the fuzzy difference-in-differences approach allows us to also make use of the different treatment intensities (due to differences in the magnitudes of the DTA revaluations) and thereby better connects the observed effects to the specific treatment of DTAs through the TCJA (Chaisemartin and D'Haultfœuille 2018).

Our results provide strong evidence that depositors react to the DTA revaluations by withdrawing deposits and the magnitude of the withdrawals increases as the amount of DTA revaluations increase. That is, depositors do not discard the mechanical revaluations of DTAs due to the TCJA. These differences persist after controlling for a large set of variables such as operating income, loan loss provisions, and regulatory capital of banks. These results support the idea that the high costs of information processing prevent depositors from fully integrating economic information like that in accrual accounting for income taxes. As a result, they react just to the bottom-line net income.

To better understand the underlying mechanisms, we explore different drivers of how the depositors react to financial accounting information. First, we condition our effect on geographic differences in the *information awareness costs* (IACs).⁴ Studies have frequently made arguments that continuously watching for the appearance of a particular firm's disclosure is costly (Blankespoor et al. 2020). The IACs refer to the costs of monitoring for this appearance (Blankespoor et al. 2019). The IACs are important to depositors' use

of income statement information because they often do not actively search for these statements but rely on local news outlets to provide earnings information. Hence, better coverage of earnings news is a prerequisite for less sophisticated depositors to become aware of banks' financial disclosures (Blankespoor et al. 2018). Appendix B provides some examples of how local media coverage makes depositors aware that a given bank has new earnings information. Local newspapers frequently cover the disclosures of local banks, even if they are privately held or are part of a listed bank holding company. In line with this argument, we find significantly higher withdrawals from depositors in areas with more local newspapers (Heese et al. 2022) and where the local media is more likely to cover earnings information (Engelberg and Parsons 2011).

Next, we explore the role of interdepositor differences in *information integration costs* (IICs). Once the depositors are aware of the existence of a disclosure signal and obtain its information, for example, by reading about it in the local media, they have to combine and refine the acquired signals to give them meaning. From the examples in Appendix B, it becomes clear that the media coverage only provides raw pieces of information that require further mental effort by the depositors to extract the relevant pieces of information. In line with Blankespoor et al. (2020), we refer to the time and mental effort that is necessary to combine the disclosed signals as IICs. Retail depositors might rationally underinvest in integrating detailed accounting information (Banerjee et al. 2020). We find the highest number of withdrawals for small, insured retail depositors who have arguably the highest net IICs. They largely fail to undo the effects of the DTA revaluations on income. We find less, but still significant, withdrawals by larger, uninsured depositors who possess more financial expertise and have higher incentives to process financial information. Our results are consistent with at least some of these larger, more sophisticated depositors failing to fully integrate accounting information. In the case of deposits that are managed by professional brokers with high financial expertise who benefit from economies of scale and scope to lower their marginal IICs, we find no evidence of withdrawals. Our results show that these professional deposit brokers correctly undo the effects of DTA revaluations on income.

In additional tests, we find higher withdrawals from depositors located in geographic areas with a lower share of occupations in financial accounting. In these areas, depositors arguably face higher IICs. Our results provide evidence that lower IACs strengthen depositors' reactions to financial information. In contrast, higher IICs prevent depositors from making use of all available information but do not diminish their reactions. Taken together, depositors' reactions to situations

with low IACs but high IICs follow heuristics that are highly correlated among depositors.

We also show that depositors' reactions have additional economic consequences. Specifically, banks pay higher interest rates on their deposits to attract new depositors after they report DTA revaluations and are less able to expand their loans.

Structural differences between banks with and without DTAs are the major concern for our empirical identification strategy. We address these concerns in a variety of ways. First, we investigate trends in deposits for banks with and without DTAs. We find common trends before and long after the DTA revaluations but not shortly following the TCJA. Consequently, depositors' reactions are unlikely to be driven by fundamentally different trends between banks with and without DTAs. Second, we add a large set of control variables as well as time and state or bank fixed effects to account for differences in banks' business models, in their business cycles, and their locations. Third, we use a matching approach to deal with the potential misspecification of the functional form (Shipman et al. 2017). Fourth, we use geographic and interdepositor differences in split samples. Our results show that depositors' reactions to DTA revaluations due to the TCJA are strongest if the costs of information acquisition are low but the costs for information integration are high. The inferences are robust across all specifications that indicates their reactions can be attributed to the TCJA's tax effects.

We contribute to the literature on the role of information processing costs (see Blankespoor et al. (2020) for a literature review) by investigating their influence on the reactions of depositors, whose incentives differ from other stakeholder groups. We show that information processing costs play an important part in understanding depositors' reaction to risk-irrelevant information. Our findings indicate that in situations where risk-irrelevant information is included in net income, the costs of becoming aware of and integrating information based on the depositors' reactions go in different directions. Specifically, if the awareness costs are low but the integration costs are high, depositors use heuristics and withdraw deposits, although the risk profile of the banks has not changed.

Furthermore, we contribute to the literature on how and why depositors react to banks' financial reporting. Consistent with the findings in Chen et al. (2022), our analyses confirm that depositors react to financial statement information. Although Chen et al. (2022) shows that the discretionary application of accounting rules has an influence on depositors' decision making, we also show that unbiased information can lead to unwarranted withdrawals of deposits due to high information processing costs. Thereby, we also add to the literature on adverse effects of bank-level transparency (Dang et al.

2017). This literature claims that too much available information about banks can be detrimental because it increases the likelihood of actions by depositors and hampers banks' ability to create money-like securities (Dang et al. 2017). We show that it is not only the overall level of transparency but also the components that are included in net income.

Finally, we contribute to the literature on the declining relevance of the income statement (see Srivastava (2023) for an overview) and the resulting debate on the general presentation and disclosure of financial information. Srivastava (2023) argues that the net income is losing relevance due to a lack of matching between revenues and expenses (Dichev and Tang 2008, Gu et al. 2021, Green et al. 2022). We show that information processing costs are an additional problem for the relevance of the income statement. The literature on the recognition of fair value changes for liabilities addresses a similar problem, but this literature deals with risk-relevant information and does not address the role of information processing costs. For example, Gaynor et al. (2011) use an experiment in which 70% of the participating CPAs misinterpreted the changes in credit risk when liabilities were valued at fair value. Govindarajan et al. (2018) proposes a supplementary item in the financial reports to increase the relevance of the financial statements that would highlight the risk-irrelevance of some items, but it seems unlikely that standard setters will implement such far reaching proposals. However, to deal with the misinterpretation of risk-irrelevant information a more modest change seems sufficient. The IASB reacted to the misinterpretation of fair value changes for liabilities by mandating the recognition of the resulting gains and losses in other comprehensive income (instead of profit and loss) that helps to prevent the misinterpretations of firm performance (Lachmann et al. 2015). Thus, standard setters can consider a similar treatment for risk-irrelevant information. We acknowledge that our results are based on regulatory call reports whose objective is not to inform depositors. However, the net income in the call reports follows the GAAP (Badertscher et al. 2018), and so we do not expect different results when using GAAP information.

2. Institutional Background and Development of Hypotheses

2.1. Tax Accrual Accounting and the 2017 TCJA

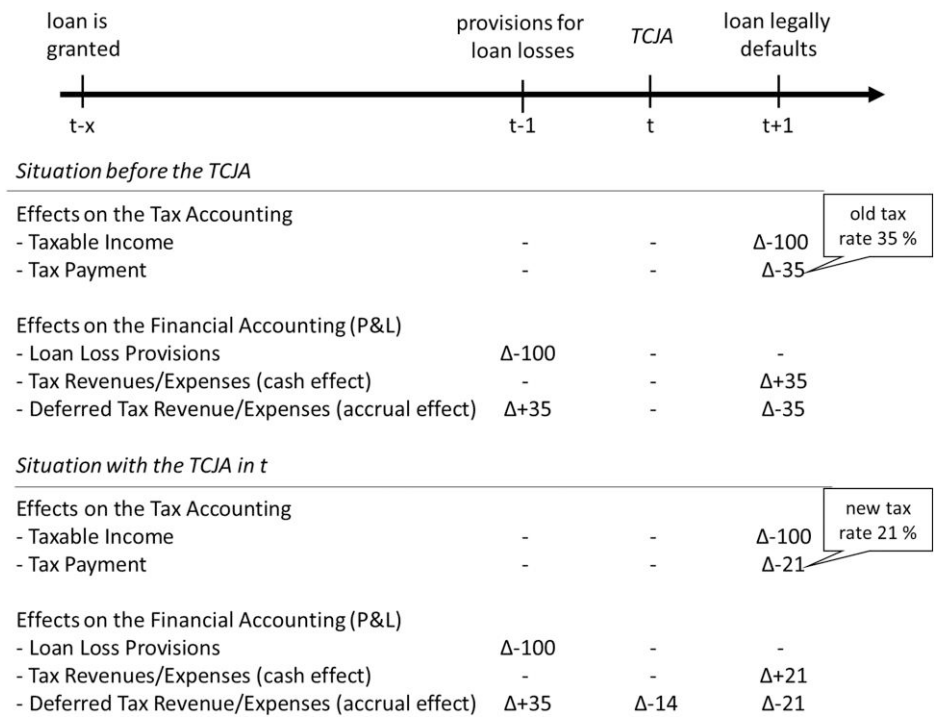
Banks frequently recognize DTAs. When the tax basis of an asset (liability) differs from its reported amount in the financial statement and that difference results in taxable income or a tax deduction on reversal, the ASC 740 requires firms to recognize tax accruals on their balance sheets. The tax accruals capture the expected tax benefits and costs in the corresponding period (Graham et al. 2012). Banks' business models often

involve temporary differences between book and tax values because tax accounting applies stricter requirements to the recognition of banks' expenses for the provision of loan losses (Hanna et al. 2019). Financial accounting recognizes the tax-reducing effect of expenses when the expenses are recognized (ASC 740). Thus, there are (deferred) tax revenues when expenses occur but a corresponding (deferred) tax expense when the tax payment happens. In addition, many banks accumulated net operating losses during the financial crisis that permitted future tax-free income (Hanna et al. 2019). DTAs measure the future tax benefits that are attributable to accounting expenses that are under tax rules not yet deductible. By contrast, deferred tax liabilities (DTLs) measure the future tax obligations that are attributable to accounting income that is not yet realized under tax rules. Therefore, the DTAs represent future tax-free income, such as the amount of the underlying temporary book-tax differences that are multiplied by the applicable corporate tax rate (Graham et al. 2012, Laux 2013).

The TCJA, which rapidly passed into law in the fourth quarter of 2017, cut the maximum corporate income tax rate from 35% to 21% for all firms. For banks with DTAs, this had an additional effect: the large tax cut led to a mechanical revaluation of DTAs (Gaertner et al. 2020). The TCJA does not affect the amount of future tax-free income, but it reduces the overall tax rate on banks' future income and consequently the value of the benefits from having tax-free income. Although the TCJA forces banks with DTAs to revalue their accruals, the underlying tax base (i.e., the revenues minus the expenses that are already recognized in the financial accounting) is unchanged and so there are no direct effects on the risk-taking of the banks.

The TCJA's effect on banks' DTAs becomes clearer when looking at a stylized example: Consider a bank that pays out a loan to one of its clients in period $t - x$ (with x greater than one). In period $t - 1$, the bank learns that the client will be unable to repay the loan and recognizes USD 100 million of loan loss provisions on its financial accounts. Because the tax accounting recognizes the costs of the defaulted loan only in period $t + 1$ when the loan legally defaults, the banks' taxable income in period $t + 1$ decreases by USD 100 million with a tax savings before the TCJA of USD 35 million (USD 100 million times the tax rate). As a consequence, there is a temporary difference between accounting profit and taxable profit in period $t - 1$ that will reverse in period $t + 1$. The financial accounting aims at allocating the tax benefits to the period of its origin (i.e., period $t - 1$) by recognizing a deferred tax asset. Specifically, the bank recognizes DTAs of USD 35 million and a corresponding (deferred) tax revenue of USD 35 million in period $t - 1$. As a result, the income tax expense in the profit and loss statement corresponds to the accounting

Figure 1. Stylized Example of the Effect of the Tax Cuts and Jobs Act on Banks’ Tax Expenses



Notes. This figure displays the stylized example discussed in Section 2. In period $t - x$, a bank pays out a loan. In period $t - 1$, the bank learns that the client will not be able to repay the loan and recognizes USD 100 million of loan loss provisions, but the tax accounting does not allow the deduction of an expense at this stage. To account for this temporary difference, the bank recognizes DTAs of USD 35 million. In period $t + 1$, the loan legally defaults, and the loss is recognized in the tax accounting. The bank de-recognizes the DTA and books a (deferred) tax expense of USD 35 million. In the lower part of the figure, the same scenario is repeated, but with the TCJA in period t .

profit. At the time the loan legally defaults in period $t + 1$, the bank also recognizes the expenses in the tax accounting, and the tax-book difference reverses. The bank de-recognizes the DTAs and books a (deferred) tax expense of USD 35 million. Figure 1 illustrates this example.

Now suppose the TCJA cuts the corporate income taxes in period t from 35% to 21%. The tax relief of the loans’ default in period $t + 1$ reduces to USD 21 million, and the bank has to adjust its DTAs by USD 14 million at the time of the TCJA, that is, in period t . The only difference from the initial situation is that the de-recognition of the DTAs occurs in two steps. Nevertheless, the after-tax income in the period of the TCJA decreases by USD 14 million because of the DTA revaluation. As this is a mechanical effect, it is possible to anticipate the revaluation based on the number of DTAs disclosed in the previous quarter’s balance sheet. Wagner et al. (2018) present empirical evidence for such an anticipation effect in the equity market: shareholders expected the revaluation and reacted as early as the law was proposed.

The example in Figure 1 shows that after the TCJA banks can deduct only 21 cents instead of 35 cents on a dollar of losses that they had previously recorded (see the

line “Tax Revenues/Expenses (cash effect)” in Figure 1 that is lower in the situation with the TCJA). In isolation, this is bad news for depositors because the bank will not be able to realize the full benefit of the tax assets they recorded. However, this disadvantage is counterbalanced by a lower tax on future profits. Banks can only realize the tax benefits from DTAs if they have sufficient profits in the future and, therefore, are not allowed to show DTAs in their balance sheet if they do not expect such profits (ASC 740-10-30-18). In our example, banks are only allowed to recognize the DTAs if they expect at least future profits of USD 100 million against which to utilize the deduction. These future profits are now taxed with 21% instead of 35%. For this reason, the lower tax on future profits at least equals the lower deductibility, and the overall effect of the tax cut on banks’ cash flows can never become negative. However, we acknowledge that it is possible that depositors deem it unlikely that the bank will generate sufficient future profits to counterbalance the lower deductibility. Only in this arguably unrealistic scenario, the DTA revaluations are bad news for depositors. Moreover, the DTA revaluation due to the TCJA does not affect the quality of its loan portfolio because the DTA revaluation due to the TCJA comes from loans generated long in

the past (period $t - x$). It also does not affect the bank's regulatory leverage because DTAs are already subtracted when calculating the regulatory capital. For our sample banks, the FDIC explicitly demands the deduction of DTAs from Tier 1 capital in §324.22(d) of the definition of regulatory capital adjustments and deductions (FDIC 2021, 78 FR 55471). Only if DTAs are realized within one year can they be part of the Tier 1 capital, but only if they account for less than 10% of that capital. Consequently, the effects of DTA revaluations on banks' regulatory capital are, at best, marginal (Hanna et al. 2019). Thereby, the authorities make the regulatory capital independent of changes in and revaluations of DTAs.⁵

One real-world example of how the DTA revaluations affect banks' income statements is provided in Appendix A. Appendix A shows that the DTA revaluations do not affect the items in the income statement

that refer to the operating expenses and revenues but show up in the tax expenses and the bottom-line net income.

To back up our claim of the risk-irrelevance for the DTA revaluations from the TCJA, we empirically investigate whether these DTA revaluations correlate with changes in banks' riskiness as measured by (a) banks' share of nonperforming loans on their loan portfolios, (b) the amount of charged-off loans as a share of the total loans, or (c) their regulatory capital. In addition, we borrow from Wagner et al. (2018) and empirically investigate whether there are abnormal equity market returns for the subsample of domestic commercial banks with listed equity around the announcement of the reports in Q4 2017, that is, once the DTA revaluations due to the TCJA affected the banks' net income as shown in Panel B of Table 1. Wagner et al. (2018) show that equity investors anticipate the implication of the

Table 1. Empirical Test on Risk Relevance of DTA Revaluations

Panel A: Correlation with key risk metrics		
	DTA revaluations from the TCJA	
	Correlation coefficient	Prob > T
Change in the NPLs _(t+1)	−0.0585	0.1020
Change in the Charge-offs _(t+1)	0.0121	0.7418
Change in the Equity ratio _(t+1)	0.0436	0.2111
Panel B: The capital market reaction around the Q4 2017 reports		
Dependent variable	(1) Three-day cumulated abnormal returns around the announcement of the quarterly reports	(2) (3)
NOL DTAs	0.004 (1.480)	
Gross DTAs		0.002 (1.469)
Net DTAs		0.002 (1.186)
Controls		
GAAP ETR	0.000 (0.153)	−0.000 (−0.419)
Ln(MV Equity)	−0.002* (−1.667)	−0.002 (−1.382)
Δ Net Interest Income	0.000* (1.759)	0.000* (1.741)
Profitability	0.001 (0.235)	0.002 (0.359)
Observations	290	290
R ²	0.055	0.054

Notes. Panel A displays the Pearson's correlations of the DTA revaluations due to the TCJA with the subsequent changes in three major proxies for banks' riskiness: the changes in the rate of (a) nonperforming loans to total loans, (b) loan charge-offs to total loans, and (c) the regulatory equity capital to total assets. Panel B reiterates the analysis from columns (1) and (2) of tables 2 and 4 in Wagner et al. (2018) for the time period when the DTA revaluations first appear in banks' financial statements. It uses a subsample of 290 banks in publicly listed bank holding companies at the beginning of 2018 to estimate the following OLS model: $CAR3D_{it} = \beta_1 DTA_{sit} + Controls + \varepsilon_{it}$. We use the three-day cumulated abnormal stock return (CAR) around the announcement of the quarterly report of Q4 2017 as the dependent variable. Following Wagner et al. (2018), NOL DTAs measure the DTAs from net operating losses scaled by the market value of equity, with both from Q3 2016. Gross (Net) DTAs cover the gross (net) amount of total DTAs scaled by the market value of equity. The control variables (GAAP ETR; Ln(MV Equity); Profitability) closely follow those in Wagner et al. (2018). In addition, we use the change in net interest income as our bank-level equivalent for revenue growth. Robust standard errors are clustered at the BHC level. Robust t statistics are displayed in round brackets.

***, **, and *Significant at the 1% level, 5% level, and 10% level, respectively.

DTA revaluations for the net income at the time the TCJA became law. Therefore, we do not expect any additional reaction from the equity market at the end of 2017 when the change in tax rate finally affects the banks' net income.

Panel A of Table 1 displays the Pearson correlations of the DTA revaluations due to the TCJA and the subsequent changes in the risk measures. Panel A shows that the DTA revaluations do not significantly correlate with higher risk levels in the coming period. We even see a negative correlation for banks' nonperforming loans that turns statistically significant at just above the 10% level ($p = 0.1012$). Looking at the capital market reactions in Panel B of Table 1, we see no statistically significant reaction from the capital market to the DTA revaluations due to the TCJA. If any, the lower bottom-line net income from the DTA revaluations is associated with positive abnormal returns, although the coefficients remain below conventional significance levels. Panel B is consistent with a lack of new risk and cash-flow relevant information from the DTA revaluations in Q4 2017 because the capital market participants anticipate these effects long before the DTA revaluations enter the financial statement.⁶ Overall, the results in Table 1 underscore the lack of risk relevance for DTA revaluations due to the TCJA.

2.2. Development of Hypotheses

2.2.1. Depositors' Reactions to Banks' Financial Information. Bank depositors act on financial information, even if they are insured against financial losses.⁷ Most of the literature uses the context of bank runs to investigate depositors' reaction (Chen et al. (2022) is a notable exception), but there are some general features in these models that are relevant to the behavior of depositors in general.

Chari and Jagannathan (1988) show that depositors are potentially misled by publicly observable signals, if only some depositors are well informed. Calomiris and Khan (1991) investigate the depositors' choice to acquire information. They highlight situations where only some depositors acquire costly signals, whereas other depositors remain rationally uninformed. Similar predictions on the role of noisy private information are made by Jacklin and Bhattacharya (1988).

The models treat information integration as a costless exercise and assume that depositors adequately understand and use the available information (Chari and Jagannathan 1988, Jacklin and Bhattacharya 1988, Calomiris and Khan 1991). In our study, we rely on insights from recent studies (see the review in Blankespoor et al. 2020) to argue that depositors do not only need to acquire a (noisy) signal but also have to invest in the integration of the information. If the information processing costs are too high, economic agents often make decisions using only a subset of partially understood

information (Blankespoor et al. 2020). These decisions are characterized as "heuristics" (Kahneman 2012), but the prevalence and consequences of using only a subset of information has been studied mainly in equity markets (Sloan 1996, Barber and Odean 2008, Lachmann et al. 2015, Blankespoor et al. 2020). However, we cannot easily generalize the depositors' reactions from these findings because they face unique economic tradeoffs.

First, demand deposit contracts are subject to the so-called "sequential service constraint" by which payments are made on a first-come, first-serve basis (Diamond and Dybvig 1983, Calomiris and Khan 1991, Chen 1999, Goldstein and Pauzner 2005). Consequently, it is beneficial for the depositors to withdraw early based on a subset of well-understood information because acquiring and integrating additional information on the bank is time consuming and puts the depositors back in the line. The deposit insurance, in theory, diminishes the incentive to withdraw deposits fast, but the FDIC openly acknowledges that many depositors are skeptical and expect a long period of time before they receive their deposit insurance payments from the FDIC. Specifically, the FDIC refers to an expected waiting period of multiple years as one of the five common misperceptions that it faces (FDIC 2014).

Second, demand deposit contracts are debt contracts. Unlike equity investors, depositors do not benefit from the gains of already healthy banks and cannot short-sell the deposit contracts to express adverse opinions. The asymmetric distribution of gains and losses, which depositors share with other debt investors, makes costly information acquisition and integration less beneficial compared with equity investors.

Third, retail depositors are, on average, financially less sophisticated and, consequently, experience higher costs of integrating financial information. At the same time, they invest only a small amount of money in deposits and often have multiple accounts at different banks.⁸ Their demand deposit contracts consequently do not allow them to leverage the information processing costs, but they bear little switching costs and other barriers to transfer the deposits to a different bank.⁹

The specific economic incentives of depositors shift the balance to making decisions based on an initially smaller subset of well-integrated information. The media dissemination of banks' disclosures allows depositors to acquire some easily digestible information like the net income with sufficiently low costs.¹⁰ In contrast, they will not integrate the more difficult to integrate details of the income statement. Managers try to avoid the negative effects of one-time charges by publishing proforma earnings that exclude the transitory items (Graham et al. 2005). However, the research has shown that the disclosure of proforma earnings

cannot fully remedy the mispricing (Rouen et al. 2021), and Lerman (2020) finds that nonprofessional investors focus strongly on earnings but not on line items or proforma earnings. Therefore, depositors might not realize that some components of earnings, such as the change in tax accruals, are risk-irrelevant and fixate on the reported net income. If the net income creates an adverse impression at first glance, depositors will react heuristically to that first, adverse impression even if it is not backed by substantial economic reality.

In summary, the literature supports the idea that depositors spend less effort on collecting and processing financial information if they are better insured, but they still react to it. The effort allocated to acquiring and integrating information, nevertheless, is limited and depends on the characteristics and incentives of depositors. Consequently, they fixate on net income and thus overreact to one-time negative earnings due to the DTA revaluations. The deposit insurance does not prevent withdrawals because depositors are unsure about timely reimbursement.

Thus, we predict that the following hypothesis (stated in alternative form) will hold.

Hypothesis 1. *Depositors withdraw deposits following (risk-irrelevant) DTA revaluations.*

2.2.2. Role of Depositors' IACs. Bank depositors first need to learn about the disclosures before they react to any information in the financial statements. In line with Blankespoor (2019) and Blankespoor et al. (2019), we refer to the cost of becoming aware of the pure existence of a disclosure as IACs. Because bank depositors are typically unsophisticated and do not directly follow banks' earnings disclosures, the media are likely to be the source of their information about them (Bushee et al. 2010). Because our sample mainly covers small and locally operated commercial banks, the local newspapers most likely disseminate the news on their earnings. For capacity reasons, local newspapers often use fully automated text to disseminate the earnings news without any major journalistic content (Blankespoor et al. 2018). These "robo" journalistic contents cover the most prominent information, such as the bottom-line net income, but might fail to capture all relevant information in earnings disclosures (Guest 2021). Appendix B provides two examples of these robo journalistic articles on local banks by their local newspapers.

We argue that depositors, on average, react more to information in income statements from geographic regions where the media more habitually covers this news. In these areas, the depositors' IACs are lower. In line with Heese et al. (2022), we argue that locally focused news publications more intensively cater to

their local audience and cover information on their local firms.¹¹ Consequently, depositors, on average, react more to financial statement information in geographic regions with more locally focused news publications. We also borrow from Engelberg and Parsons (2011) to compare depositors' reactions in regions with higher versus lower media coverage of local firms' earnings information. In both tests, we argue that the media coverage mainly affects the information awareness but not the IICs. It is possible that the news coverage of earnings helps the depositors to better understand the one-time character of the DTA revaluations due to the TCJA and why this news is not bad for the bank. Overall, better media coverage makes bank depositors better aware of the existence of banks' financial information. We predict that the following hypothesis will hold.

Hypothesis 2. *Depositors react more strongly to banks' DTA revaluations if their IACs are lower.*

2.2.3. Role of Depositors' IICs. Once depositors are aware of financial information, the costs of integrating it determine whether and how they incorporate the information into their reactions. Therefore, we expect cross-sectional heterogeneity in depositors' reactions that is conditional on their level of financial expertise (Bushman et al. 1996) and their economies of scale for information (Blankespoor et al. 2020). Cross-sectional analyses in equity markets find more severe mispricing of accruals in situations with more nonprofessional investors who face higher IICs (Collins et al. 2003, Pincus et al. 2007, Blankespoor 2019).

We expect that differences in the IICs also determine how the information in the financial statement influences depositors' reactions, especially if the information is highly complex. We start by measuring the IICs for different types of depositors. There are two conflicting effects that potentially affect their reactions. First, deposit insurance should make both small retail and brokered depositors react less to any kind of financial information, whereas uninsured depositors should react to any type of relevant financial information. Second, the difference in the IICs should cause different depositor types to react more strongly. We expect the strongest reaction from small but insured retail depositors who show high net IICs because their average level of expertise is low, and they do not benefit from economies of scale. We expect weaker reactions from uninsured depositors who show lower IICs due to more financial expertise and better economies of scale. We expect the weakest reactions to come from brokered deposits that are managed by professional managers with the highest level of expertise and the strongest benefits from economies of scale and scope. Deposit brokers are highly regulated, and

their employees typically undergo formal business education to show a strong expertise in the banking sector to satisfy regulatory requirements.¹² The difference between brokered and small retail depositors thereby lies mainly in the different IICs because both enjoy the same status with deposit insurance.

Tax accruals are among the most complex accounting topics (Balakrishnan et al. 2019), and there is empirical evidence that even sophisticated capital market participants, such as some analysts (Chen and Koester 2021), fail to fully understand the effect of the TCJA on firms’ earnings. In contrast to Chen and Koester (2021), our identification strategy does not require a full understanding of the earnings consequences of the TCJA. Rather, it rests on the assumption that the DTA revaluations are identified as a one-time effect that does not heavily affect the long-term financial health of the banks. For our cross-sectional tests, we only assume that there are differences in the propensity to draw appropriate conclusions from these one-time DTA revaluations between brokered, uninsured, and insured retail depositors. We do not assume that all brokered depositors properly integrate the information. Moreover, the empirical evidence from Wagner et al. (2018) indicates that equity investors were able to properly integrate the information about the TCJA and its effects on tax accruals. The findings by Wagner et al. (2018) make us confident that the more sophisticated depositors, especially the deposit brokers but also some uninsured depositors, were able to properly integrate the information of DTA revaluations from the TCJA.

As alternatives, we first measure depositors’ marginal IICs with a geographic proxy for their occupations. We argue that the general business expertise and the specific accounting expertise is closely linked to their occupations. We start by splitting our sample based on whether their metropolitan statistical area belongs to the upper or lower half based on the share of employees with a “Business and Financial Operations Occupations” (Occupation Code: 13-0000). In a more specific test, we split our sample into the shares of employees that have a direct financial accounting or a tax focus. These are occupations classified as accountants, auditors, actuaries, bookkeeping, accounting, auditing clerks, economists, financial and investment analysts, financial risk specialists, financial specialists, tax examiners and collectors, revenue agents, and tax preparers. We assume that people from these professions do not only use their expertise for their own economic benefit but also share their knowledge and their judgements with their families, friends, and their local communities. Because banks’ choice of location is exogenous to their financial reporting, at least in the short term, these geographical proxies mitigate the

Figure 2. Summary of the Predicted Depositor Reactions

		IICs	
		Low	High
IACs	High	No reaction (depositors are not aware of losses)	
	Low	No reaction (depositors adjust losses)	Withdrawal of deposits

Note. This figure displays a summary of the depositors’ predicted reactions conditional on the level of the information acquisition costs (IACs) and the information integration costs (IICs).

concerns that the amount of DTAs prior to the TCJA and the type of deposits correlate with another unobserved variable.

Overall, IICs might inhibit the depositors’ use of the full information content if the financial information is complex. Therefore, we predict that the following hypothesis will hold.

Hypothesis 3. *Depositors react more strongly to banks’ DTA revaluations if their IICs are higher.*

In Figure 2, we summarize the predictions of our cross-sectional tests. We expect a withdrawal of deposits if the IACs are low but the IICs are high. In contrast, if the IACs are high, we expect no reaction because the depositors are not aware of the loss. If the IACs and the IICs are low, we expect that the depositors understand the risk-irrelevance of the DTA revaluations and do not react.

3. Data and Research Design
3.1. Sample and Descriptive Statistics

We obtain the data on the bank-level variables from financial statements in the quarterly call reports as disseminated by the Federal Financial Institutions Examination Council. A priori, it is unclear whether depositors react to financial information from the top bank holding company (BHC) that potentially operates under a different name or from the subsidiary commercial bank alone. Because the breakdown of depositors is unavailable at the BHC level, we rely on financial information from the commercial banks. This information might be a noisy proxy if depositors also incorporate financial information from the BHC. Our sample covers January 2015 through December 2019; that is, two years and three quarters before the TCJA, one quarter with DTA revaluations, and two years after the TCJA. Because our dependent variables come from one period ahead, we collect financial information up to the first quarter of

Table 2. Descriptive Statistics

Variable	N	Mean	Standard deviation	p5	p25	p50	p75	p95
Total Deposits	39,743	0.84587	0.06473	0.72861	0.80902	0.85239	0.88855	0.94092
Insured Deposits	39,743	0.55168	0.12270	0.31750	0.47730	0.56605	0.64024	0.73043
Uninsured Deposits	39,743	0.29566	0.12359	0.12524	0.20785	0.27470	0.36561	0.53630
Brokered Deposits	39,743	0.03445	0.05754	0.00000	0.00000	0.00664	0.04743	0.15258
Change in Total Deposits	39,743	0.01455	0.04005	−0.03680	−0.00762	0.00919	0.02941	0.08029
Change in Insured Deposits	39,743	0.00588	0.02436	−0.02306	−0.00576	0.00267	0.01265	0.04430
Change in Uninsured Deposits	39,743	0.00867	0.03021	−0.03502	−0.00774	0.00554	0.02179	0.06155
Change in Brokered Deposits	39,743	0.00099	0.01182	−0.01434	0.00000	0.00000	0.00002	0.02163
Deposit Rate	39,743	0.00103	0.00080	0.00025	0.00051	0.00078	0.00128	0.00276
Level of Net DTAs	39,743	0.00245	0.00421	−0.00158	0.00000	0.00116	0.00387	0.01021
Change in Net DTAs	39,743	−0.00009	0.00160	−0.00142	−0.00032	0.00000	0.00007	0.00103
DTA Revaluations _i × Period _{Q4 2017}	1,263	0.00119	0.00173	0.00000	0.00000	0.00073	0.00180	0.00386
Operating Income before Taxes	39,743	0.01330	0.00711	0.00273	0.00934	0.01329	0.01705	0.02447
Income Taxes	39,743	0.00275	0.00290	0.00000	0.00007	0.00247	0.00438	0.00775
Loan Loss Provisions	39,743	0.00045	0.00089	−0.00006	0.00000	0.00027	0.00059	0.00170
Loan Charge-Offs	39,743	0.03972	0.07022	0.00000	0.00161	0.01379	0.04393	0.17765
Size (LnTA)	39,743	13.182	1.269	11.667	12.193	12.899	13.904	15.695
Equity Ratio	39,743	0.11267	0.02390	0.08318	0.09662	0.10836	0.12362	0.15822
Loan-to-Asset Ratio	39,743	0.71749	0.10558	0.53733	0.63802	0.72297	0.79728	0.88568
Nonperforming Loans	39,743	0.00770	0.00894	0.00000	0.00185	0.00492	0.01016	0.02529

Note. The table presents the descriptive quarterly statistics for the sample of the 3,868 commercial banks from 2015 to 2019.

2020. Because the COVID-19 crisis most likely affects any period after this quarter, we do not consider data from later periods.

We only consider financial information from domestic U.S. commercial banks (filing type: 041) to avoid any competing influences from the TCJA’s changes to the tax rules for international subsidiaries. Furthermore, we exclude banks with total assets of less than USD 1 billion (filing type: 051). These banks face less regulatory supervision and have less sophisticated financial statements with missing information for some of the variables of interest. All our commercial banks end their fiscal year in December, and all quarters refer to calendar quarters. Our main sample covers 39,743 quarterly observations of 3,868 commercial banks.

Table 2 shows that the sample banks on average fund 84.587% of their assets with deposits. The total deposits break down to 55.168% for *Insured Deposits* and 29.566% for *Uninsured Deposits*. *Brokered Deposits*, which are part of the insured deposits, only account for 3.445% of the banks’ assets. Following Chen et al. (2022), we first use the change in total deposits and the change in the different deposit types as the dependent variables. These changes are nicely distributed around zero with small mean values (between 1.455% and 0.099%) but with standard deviations between 4.005% and 1.182%. In our second specification, we also use the levels of total deposits and the levels in the different deposit types as the dependent variables in combination with bank fixed effects. Banks’ net DTAs (DTAs net of DTLs) account for 0.245% of the total assets with a standard deviation of 0.421%.

Although these numbers may seem small, banks are highly leveraged, and a comparison of the effect of net DTAs to banks’ operating income yields more meaningful insights. One standard deviation in net DTAs can explain about 59.212% of the standard deviation in banks’ operating income to total assets. Furthermore, as indicated by the newspaper articles in Appendix B, in many cases the large one-time DTA revaluation around the TCJA easily exceeds the banks’ pretax income. The operating income before taxes to assets shows values around 1.330% and the return on equity of around 11% (unreported). These values and the control variables are similar to other studies on commercial banks (Beatty and Liao 2014, Bushman and Williams 2015, Chen et al. 2022). Online Appendix A separately presents the descriptive statistics for the subsample with and without DTA revaluations. The evidence indicates that both samples are similar in most dimensions.

3.2. Research Design

3.2.1. Main Regression Model. In the base model, we analyze whether bank depositors react to banks’ DTA revaluations due to the TCJA. Therefore, we regress the subsequent change in the deposit levels on the amount of DTA revaluations with state fixed effects. Alternatively, we use the level of the deposits scaled by total assets as our dependent variables with bank fixed effects. All specifications include quarter fixed effects. We interact the amount of DTA revaluations with the indicator variable for each period at and after the TCJA to explore how the treatment effects persist

over time:

$$\begin{aligned}
 (\text{Change in/Level of}) \text{ DepVar}_{i,t+1} = & \\
 & \sum_{\tau=Q4\ 2017}^{2019} \beta_{\tau} \text{Period}_{\tau} * \text{DTA Revaluations}_i \\
 & + \text{DTA Revaluations}_i \\
 & + \text{Income Taxes}_{i,t} + \text{Business Model Controls}_{i,t} \\
 & + \text{Quarter FE}_t + \text{State/Bank FE}_i, \quad (1)
 \end{aligned}$$

where the dependent variable *Change in (Level of) DepVar_{i, t+1}* is either the *Change in (Level of) Total Deposits* of bank *i* at time *t + 1*, the *Change in (the Level of) Insured Deposits* of deposits up to USD 250,000 per account that are insured by the FDIC, the *Change in (the Level of) Uninsured Deposits* of all deposits that exceed USD 250,000 per deposit account, or the *Change in (the Level of) Brokered Deposits* of those deposits that are managed by a professional deposit broker. *Period_τ* is a set of indicator variables that equal one for each period at and after the TCJA, where *τ* represents either Q4 2017, Q1 2018, Q2 2018, Q3 2018, Q4 2018, or 2019. We refrain from using separate indicator variables for each quarter of 2019 to enhance the readability of our tables. The base terms of the indicator variables are captured by the quarter fixed effects. *DTA Revaluations_i* captures the treatment's intensity, that is, the amount of DTA revaluations that a bank experienced during the TCJA. We use the net DTAs from September (Q3) of 2017 multiplied by the change in the tax rate if the banks reported net DTAs in the third quarter of 2017 and zero otherwise. *Income Taxes_{i,t}* captures the income taxes reported in a specific quarter. All continuous variables are scaled by the banks' total assets.

We control for a broad set of variables to capture different time-varying characteristics of the banks' business model. First, we control for the level of net DTAs scaled by total assets. Second, we control for bank profitability (*operating income before taxes to total assets*), loan portfolio characteristics (*loan loss provisions to current gross loans*, *loan charge-offs to current loan loss allowances*, *loan-to-asset ratio*, and *nonperforming loans to current gross loans*), bank size (*ln of total assets*), and bank leverage (*equity ratio to total assets*). Appendix C displays a list of all variables with their descriptions and sources.

To test for the influence of the DTA revaluations due to the TCJA on potential changes in deposits, we use the interaction of *DTA Revaluations_i × Period_τ* in each period *τ* at and after the TCJA, starting in *τ = Q4 2017* for bank *i*. The variable *DTA Revaluations_i* is a bank-specific but time-invariant variable. In line with Hypotheses 1–3, we predict that the coefficient for the interaction term of the *Period_{τ=Q4 2017}* with *DTA Revaluations_i* to be negative.

That is, higher revaluations are associated with a stronger reduction in deposits.

The posttreatment effects of the DTA revaluations are captured by the interaction term *DTA Revaluations_i × Period_τ* with *τ* equal to either Q1 2018, Q2 2018, Q3 2018, Q4 2018, or 2019. These interactions capture the development of the treatment effect in the quarters directly after the TCJA through the year 2019. We do not predict a sign for these coefficients. There are different possibilities for how the effect evolves over time. On the one hand, it is possible that the deposit levels quickly reverse to their long-term mean once the bank announces earnings for the following quarter. One plausible explanation for a fast reversal is the strong short-term memory of depositors who discard old information easily. On the other hand, we possibly see a protracted response from the depositors to the earnings shocks from the DTA revaluations. One possible explanation for that response is that some depositors shy away from banks for a longer period of time and see a bank with this earnings shock as having a stigma attached to it. In addition, other depositors might not only react to the earnings shocks, but also to the reaction by other depositors to the information, which yields some kind of information cascade as discussed by Chari and Jagannathan (1988) and Calomiris and Khan (1991). Another possible explanation goes back to the increased funding costs for banks. If they need to increase deposit rates to keep and reattract deposits after a negative earnings shock from the DTA revaluations, the demand for deposits will become less attractive to the bank and that demand will decrease. Because this demand comes with some delay, the bank-demand explanation would yield protracted effects that come with some delay.

Although most banks reported net DTAs, the TCJA positively affected the few banks that reported net DTLs. We do not capture the positive revaluations for our empirical analysis and set *DTA Revaluations_i* to zero because, from a theoretical perspective, the expectations regarding the depositors' reactions are clearer for negative earnings shocks. Nevertheless, additional tests show that our results remain qualitatively the same if we also capture positive earnings shocks.

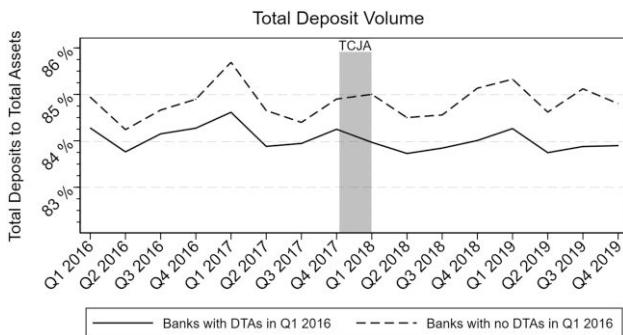
The quarter fixed effects capture the overall economic environment in each period and control for the influence of the business cycle on the overall deposit flows. If the dependent variable reflects changes in deposits, we use state fixed effects to capture the differences in the economic environment of the different states and add control variables in levels. We refrain from using bank fixed effects if the dependent variable reflects the changes in deposits. When using the levels of deposits as dependent variables, we rely on bank fixed effects instead of state fixed effects and add control variables for the levels of items in the income

statement. Because the control variables for these items are fairly stable over time and are therefore captured by the bank fixed effects, we add the changes in these variables as more controls. In all specifications of the model, we estimate the robust standard errors clustered at the bank level.

3.2.2. Common Trend. For the difference-in-differences analysis, the common trend in the variables of interest in the quarters before and after the TCJA but not during the change are essential to rule out different time trends as competing explanations. Figure 3 displays the average levels of total deposits over time. For Figure 3, we split the sample into banks with and without DTAs in the first quarter of 2016 because at this point in time, banks were very unlikely to have expected the tax cuts to happen. Therefore, we argue that the DTAs in the first quarter of 2016 are most likely exogenous to the TCJA. To rule out any distorting effects from our unbalanced sample in Figure 3, we only consider banks with no more than three quarters of missing observations. The missing observations most often are from mergers or reorganizations of banking branches.

Evidence from Figure 3 strongly supports a common trend in the deposit levels before the TCJA. We see a strong divergence at the time of the TCJA (gray area). In the periods after the TCJA, the trends become common again, although at different levels. In unreported tests, we find similar patterns for the insured as well as uninsured deposits while we do not find any changes after the TCJA for the brokered deposits. The change in trends supports our empirical identification strategy. It alleviates concerns that an alternative long-term trend can potentially cause our empirical results.

Figure 3. Common Trends in the Levels of Total Deposits

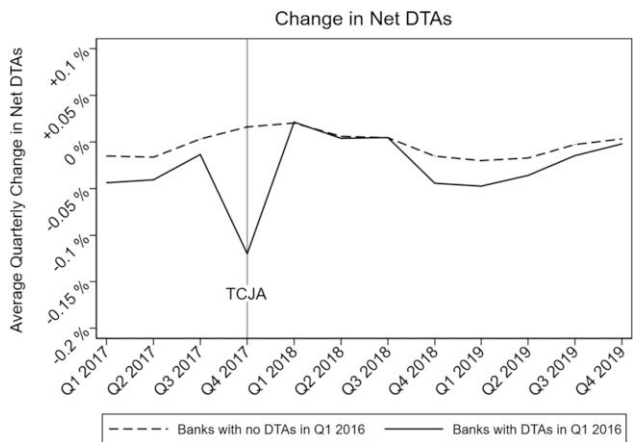


Notes. Figure 3 depicts the quarterly averages in the share of total deposits to total assets in the solid black line for the subsample of banks with net DTAs in the first quarter of 2016, that is, before the TCJA was expected. The dotted and gray line represents the subsample of banks without net DTAs in the first quarter of 2016. The gray area depicts the period between the end of Q4 2017 and the end of Q1 2018. In this period, the depositors learned about banks' earnings that were affected by the DTA revaluations due to the TCJA and were expected to react to it.

Figure 3 shows similar trends between affected and unaffected banks in the post-TCJA period but the differences in the deposit levels after the TCJA shock are persistent. This is surprising because it indicates that the depositors do not readily return their deposits to the bank from which they withdrew them. The evidence is consistent with different possible explanations, such as a shift in the investment to other asset classes or an attention effect (Barber and Odean 2008) that differs between the depositors' decision to withdraw and to return. In line with the first explanation, we see small increases in the investments in money market funds and debt securities during the TCJA quarter, and to a much smaller magnitude, an increase in corporate equity and mutual fund shares. Second, Barber and Odean (2008) have found differences in investors' attention between firms that they are invested in and other firms. The attention effect postulates that investors more closely follow the financial performance of the firms they are currently invested in and sell their investments in cases of bad performance. At the same time, they only use a subset of firms in their search for new investment opportunities (Barber and Odean 2008). Nevertheless, more empirical evidence, such as deposit account data, is needed to better understand where the deposits go and why they do not return. Because we do not have access to this kind of detailed information, the answer to this question, unfortunately, remains outside the scope of this study.

For the changes in net DTAs in Figure 4, there clearly is a common trend before the fourth quarter of 2017.

Figure 4. Common Trends in the Change in Net Deferred Tax Assets



Notes. Figure 4 depicts the quarterly averages in the changes of net DTAs from the previous quarter in the solid black line for the subsample of banks with net DTAs in the first quarter of 2016, that is, before the TCJA was expected. It further depicts the quarterly averages in the changes of net DTAs from the previous quarter with the dotted and gray line for the subsample of banks without net DTAs in the first quarter of 2016. The gray line depicts the period Q4 2017. In this period, banks' earnings were affected by the DTA revaluations from the TCJA.

Because one group by construction had no net DTAs in the first quarter of 2016, the figure starts showing the common trend at the beginning of 2017. Figure 4 shows the changes in banks' net DTAs that arguably explain the difference in deposit levels: We see large one-time decreases in banks' net DTAs at the time of the TCJA. Consequently, the graphical evidence from Figures 3 and 4 is the first to support Hypothesis 1.

4. Empirical Results

4.1. Univariate Differences

Table 3 displays the univariate tests on the changes in the overall deposit levels before, at, and after the TCJA. In Panel A, we approximate whether a bank reports DTAs by the first quarter of 2016 to avoid any reverse causality on its DTA reporting after the TCJA. Panel B shows the differences based on the treatment's intensity. We split the sample into quartiles depending on the amount of DTA revaluations due to the TCJA but exclude banks with net DTLs around the TCJA.¹³ Below and next to the columns,

we report Wald tests on the differences between the averages.

Panel A of Table 3 even shows a small but statistically nonsignificant increase in the average deposit level in the quarter after the TCJA for banks that do not have any DTAs that reverses in the following quarters. These banks' earnings are not adversely affected by DTA revaluations from the TCJA. In contrast, we find a significant reduction in the deposit levels for the banks with DTAs. This reduction in the level of deposits is the strongest (coefficient difference = -0.0016) when the financial statement information from Q4 2017 was published and further decreases by another 0.0009 in the subsequent quarters. Looking at the Wald test results, we find a significant reduction between the pre- and post-TCJA subperiods. The tests for Q4 2017 remain statistically nonsignificant that might be related to power issues in the test. At the same time, the banks with DTAs had already reported lower levels of deposits before the TCJA. This difference increases further at and after the TCJA.

Table 3. Univariate Evidence on Banks' Deposits, DTA Revaluations, and the TCJA

Panel A									
	(1)		(2)		(3)		Differences		
	Until Q3 2017		Q4 2017		From Q1 2018		(1) vs. (2) Difference	(2) vs. (3) Difference	(1) vs. (3) Difference
	N	Value	N	Value	N	Value			
No DTAs in Q1 2016	4,390	0.8485	373	0.8513	2,851	0.8488	0.0029	-0.0026	0.0003
With DTAs in Q1 2016	8,148	0.8417	686	0.8401	5,080	0.8392	-0.0016	-0.0009	-0.0026**
	Difference		Difference		Difference				
Difference	-0.007	0.000***	-0.011	0.007***	-0.010	0.000***			
Panel B									
By rank of DTAs at the TCJA	(1)		(2)		(3)		Differences		
	Until Q3 2017		Q4 2017		From Q1 2018		(1) vs. (2) Difference	(2) vs. (3) Difference	(1) vs. (3) Difference
	N	Value	N	Value	N	Value			
(1) 0%–25%	2,129	0.8468	180	0.8470	1,352	0.8482	0.0001	0.0012	0.0014
(2) 25%–50%	2,142	0.8536	181	0.8509	1,348	0.8494	-0.0027	-0.0015	-0.0042**
(3) 50%–75%	2,135	0.8385	179	0.8298	1,329	0.8325	-0.0086	0.0027	-0.0060**
(4) 75%–100%	2,146	0.8292	181	0.8310	1,329	0.8275	0.0018	-0.0035	-0.0017
	Difference		Difference		Difference				
Wald test	Difference		Difference		Difference				
(1) vs. (2)	0.008		0.018		0.009				
(1) vs. (3)	-0.017		-0.016		-0.001				
(1) vs. (4)	-0.016		-0.021		-0.005				

Notes. Panel A displays the average levels of total deposits in period $t + 1$ scaled by total assets before and after the Tax Cuts and Jobs Act. Thereby, it presents the averages for the banks with and without DTAs in Q1 2016, which is a time when the TCJA was unpredictable. Because we are interested in the depositors' reaction to the financial statement information, the quarter Q4 2017 refers to that reaction to the Q4 2017 financial information which takes place after the information was published in Q1 2018. Panel B displays the average levels of total deposits in period $t + 1$ scaled by total assets before and after the TCJA. Thereby, it presents only the banks with DTAs in the third quarter of 2017 that is just before the enactment of the TCJA. It captures the treatment intensity of the TCJA on DTAs for treated banks only. To avoid any distortions from changes in the sample's composition, the averages contain only the banks with no more than three quarters of missing observations.

***, **, and *Significant at the 1% level, 5% level, and 10% level, respectively.

Panel B of Table 3 indicates whether these differences around the TCJA vary because of the treatment's intensity. Before the TCJA, we see that banks with fewer DTA revaluations already had higher levels of deposits with banks just below the median having the largest shares of deposits. For the banks in the bottom quartile of the DTA revaluation distribution in line (1), we find no significant changes in the average deposit levels before, at, and after the TCJA. For the banks in the second and third quartile, we find a significant decrease in deposits. Looking at the differences, we find that most of the reduction in deposit levels happened in the first quarter after the TCJA when depositors became aware of the net income shock from the DTA revaluations for the first time. For the banks in the top quartile of the DTA revaluation distribution in line (4), we find no significant changes in the average total deposit levels before, at, and after the TCJA. Additional tests show that those banks experience a significant decrease in their insured retail deposits (coefficient: -0.0168 ; $p=0.093$) after the announcement of the quarterly reports of Q4 2017 but manage to overcompensate for the outflows of insured retail deposits with inflows of uninsured deposits. In untabulated analysis, we identify a potential channel of how those heavily affected banks attracted the uninsured depositors. We see a substantial increase in the deposit rates that the banks in the fourth quartile pay to their depositors. The persistent effect associated with the TCJA beyond the period when the loss is reported appears surprising at the first glance because inattentive investors might ignore lagged information. Nevertheless, it is consistent with different mutually nonexclusive explanations, such as investor wake-up calls or heuristic depositor reactions to the disclosure of the withdrawals by other depositors, that is, herding. Overall, the results in Table 3 show that the average level of deposits decreases around the TCJA only for the banks with DTAs prior to the TCJA. This reduction is strongest for banks with above median levels of DTAs and is not limited to the most extreme DTA users that had already shown the lowest deposit levels before the TCJA.

4.2. Depositors' Reactions to the DTA Revaluations

Table 4 provides multivariate evidence on whether DTA revaluations due to the TCJA trigger any changes in banks' deposits. We rely on the interaction term $DTA\ Revaluations_i \times Period_{Q4\ 2017}$ to explore whether the amount of DTA revaluations show some immediate effects on the levels of (changes in) bank deposits after the TCJA. The interaction term $DTA\ Revaluations_i \times Period_\tau$ with τ equal to either Q1 2018, Q2 2018, Q3 2018, Q4 2018, or 2019 show whether the DTA revaluations trigger effects on bank deposits in the posttreatment periods.

They differentiate as to whether any treatment effect reverses quickly or whether there is a protracted response by the depositors to the earnings shocks from the DTA revaluations. In the uneven columns, we use the change in bank deposits as our dependent variable and in the even columns the level of deposits.

In columns (1) and (2), we find a highly negative relation between the DTA revaluations due to the TCJA and the (change in the) deposit levels. In column (1), we find a statistically significant and negative effect on the change in deposits for the banks that is conditional on the earnings shock from the DTA revaluations. In column (2), we find a very similar statistically significant and negative coefficient when using the level of deposits and bank fixed effects. The effect size is also economically meaningful. With a regression coefficient of -2.1868 (-1.9429) in column (1) [column (2)] and an average DTA revaluation due to the TCJA scaled by total assets of 0.00119 , the DTA revaluations due to the TCJA explain 17.88% [15.89%] of the typical changes in total deposits.¹⁴ Therefore, the effect is economically significant and supports Hypothesis 1.

To alleviate concerns about the misspecification of the functional form (Shipman et al. 2017), we also report the results for a propensity score matched sample. Therefore, we match banks by their size, equity ratio, loan-to-asset ratio, lagged share of deposits, and operating income before taxes in the fourth quarter of 2017 when the TCJA went into effect and only keep the time series of banks with a properly matched control firm. We use one-to-one nearest neighbor matching with a caliper distance of smaller than 0.03 . Thereby our sample of matched banks substantially decreases to only $18,118$ observations from $1,059$ unique banks. The results in columns (3) and (4) are similar to those in columns (1) and (2). They show that the coefficient sizes stay similar and remain statistically significant at conventional levels. Overall, we find no indication that our results suffer from a misspecification.

Looking at the posttreatment effects, in column (1) we find some indication of a partial reversal in the short term, although the coefficients for $DTA\ Revaluations_i \times Period_{Q1\ 2018}$ and $DTA\ Revaluations_i \times Period_{Q2\ 2018}$ remain statistically nonsignificant and economically smaller than the main effect. After the partial reversal, we see some further decline that indicates some long-term effect on the banks' business models. The further deterioration turns statistically nonsignificant when using the level of deposits in columns (2) and (4) and bank fixed effects or when using propensity score matching in columns (3) and (4).

All control variables show the predicted signs. Banks' level of income taxes shows a positive coefficient on average over the sample period when considering the change specifications. Because banks with higher (lower) income

Table 4. Change in Different Deposit Types, Banks' DTA Revaluations, and the TCJA

Dependent variable: <i>Total deposits</i> _(t+1)	(1)	(2)	(3)	(4)
	Full sample		With PSM	
	In changes	In levels	In changes	In levels
TCJA: Treatment quarter				
<i>DTA Revaluations</i> _i × <i>Period</i> _{Q4 2017}	−2.1868*** (−3.4371)	−1.9429** (−2.3167)	−1.6164* (−1.7924)	−2.4892** (−2.1701)
Posttreatment effects				
<i>DTA Revaluations</i> _i × <i>Period</i> _{Q1 2018}	0.6958 (1.0746)	−0.0795 (−0.0849)	1.3332 (1.5467)	−1.0601 (−0.8926)
<i>DTA Revaluations</i> _i × <i>Period</i> _{Q2 2018}	0.4782 (0.7039)	1.0590 (1.1299)	1.1647 (1.2845)	0.2925 (0.1978)
<i>DTA Revaluations</i> _i × <i>Period</i> _{Q3 2018}	−1.4475** (−2.2675)	−0.2725 (−0.2602)	−0.3622 (−0.4283)	−1.1950 (−0.7459)
<i>DTA Revaluations</i> _i × <i>Period</i> _{Q4 2018}	−1.1693* (−1.7189)	−0.9949 (−0.8990)	−0.4116 (−0.4299)	−2.0942 (−1.3323)
<i>DTA Revaluations</i> _i × <i>Period</i> ₂₀₁₉	−1.3323*** (−3.3566)	−1.0899 (−0.9731)	−0.4785 (−1.0437)	−2.4962* (−1.6612)
Controls				
<i>Income Taxes</i> _{i,t}	0.3552*** (3.2791)	0.2176 (0.8324)	0.4301*** (2.5944)	−0.1489 (−0.3796)
<i>Level of Net DTAs</i> _{i,t}	0.0296 (0.4137)	−0.2355 (−0.9042)	0.2313 (1.3611)	−0.4054 (−1.0779)
<i>DTA Revaluations</i> _i	0.3644 (1.6220)	in fixed effects	−0.4835 (−1.1556)	in fixed effects
<i>Operating Income before Taxes</i> _{i,t}	0.0298 (0.6059)	−0.1432 (−1.5149)	−0.0087 (−0.1095)	0.0140 (0.0936)
<i>Size (in levels/ changes)</i> _{i,t}	0.0010*** (4.2911)	−0.2184*** (−20.7942)	0.0011*** (3.4232)	−0.1762*** (−10.3492)
<i>Equity Ratio (in levels/changes)</i> _{i,t}	0.0328*** (2.8949)	0.0371 (0.6973)	0.0233 (1.2995)	−0.0484 (−0.6116)
<i>Loan-to-Asset Ratio (in levels/changes)</i> _{i,t}	0.0576*** (24.3718)	0.1185*** (12.1905)	0.0588*** (16.0186)	0.1093*** (7.1664)
<i>Nonperforming Loans (in levels/changes)</i> _{i,t}	−0.2802*** (−10.6873)	−0.0167 (−0.3087)	−0.2040*** (−5.1434)	−0.0220 (−0.3025)
<i>Loan Loss Provisions (in levels)</i> _{i,t}	1.5690*** (4.1831)	−0.3031 (−0.6936)	1.6013*** (2.9548)	−0.3909 (−0.5453)
<i>Loan Charge-Offs (in levels)</i> _{i,t}	−0.0234*** (−6.0229)	−0.0077* (−1.8547)	−0.0279*** (−4.7055)	−0.0142** (−2.1961)
State and quarter fixed effects	Yes		Yes	
Bank and quarter fixed effects		Yes		Yes
Observations	39,743	35,137	18,118	16,844
R ² (within)	0.0627	0.0461	0.0601	0.0368
F	50.48	22.98	20.68	10.09

Notes. The table displays the estimation results from the following OLS panel model.

$$\begin{aligned}
 \text{DepVar}_{i,t+1} = & \sum_{\tau=Q4\ 2017}^{2019} \beta_{\tau} \text{Period}_{\tau} * \text{DTA Revaluations}_i + \text{DTA Revaluations}_i + \text{Income Taxes}_{i,t} + \text{Business Model Controls}_{i,t} + \text{Quarter FE}_t \\
 & + \text{State/Bank FE}_i + \varepsilon_{i,t}
 \end{aligned}$$

The changes in the (level of) banks' total deposits_(t+1) serve as the dependent variable in the uneven (even) columns. The quarter and state (bank) fixed effects are included in the uneven (even) specifications. In columns (3) and (4), we report the results for a propensity matched sample based on size, equity ratio, deposit levels, loan-to-asset ratio, and operating income before taxes from the TCJA. The sample contains all (only the matched) US commercial banks over the period from 2015 to 2019. Robust standard errors are clustered at the bank level. Robust *t* statistics are displayed in parentheses.

***, **, and *Significant at the 1% level, 5% level, and 10% level, respectively.

pay higher (lower) income taxes, the results from banks' income taxes point to higher deposit levels in more profitable banks. The level of DTAs appears unrelated to banks' change in deposit levels. The deposits increase at

safer and larger banks as well as at the banks that focus on loans. Depositors appear to prefer banks with fewer nonperforming loans, more loan loss provisions, and fewer loan charge-offs.

4.3. Geographic Variation in Depositors’ Reactions Conditional on Their Awareness Costs

Table 5 shows whether DTA revaluations due to the TCJA trigger stronger reactions from depositors in areas with locally focused news publications and if the local media more frequently disseminates news on earnings information to the banks’ depositors. Both arguably reduce the depositors’ IACs. We borrow from

Heese et al. (2022) who use data from the database at the UNC Hussman School of Journalism and Media to explore the branch-level consequences of local media coverage and from Engelberg and Parsons (2011) who explore the causal effect of media on financial markets. In Table 5, we first condition our effect of DTA revaluations due to the TCJA on the deposit levels to the existence of locally focused, daily newspapers in 2017 that are covered in the UNC’s database (columns (1)

Table 5. Depositors’ Reactions and Information Awareness Costs

Panel A: Multivariate regression results								
Dependent variable: <i>Total deposits</i> _(t+1)	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Presence of local newspaper				Share of local earnings coverage			
	With local newspaper		No local newspaper		Above median		Below median	
	In changes	In levels	In changes	In levels	In changes	In levels	In changes	In levels
Treatment quarter								
<i>DTA Revaluations</i> _{<i>i</i>} × <i>Period</i> _{Q4 2017}	−3.3348*** (−3.5525)	−3.3069*** (−2.5922)	−0.5022 (−0.4136)	−0.7521 (−0.5102)	−3.7785*** (−3.3255)	−3.9168** (−2.4507)	−1.2164 (−1.2364)	−1.4525 (−1.1677)
Posttreatment effects								
<i>DTA Revaluations</i> _{<i>i</i>} × <i>Period</i> _{Q1 2018}	1.2033 (1.1173)	−0.1401 (−0.0972)	0.0568 (0.0492)	−0.1090 (−0.0708)	−0.0963 (−0.1042)	−1.2076 (−0.6714)	1.3445 (1.1002)	0.5230 (0.3788)
<i>DTA Revaluations</i> _{<i>i</i>} × <i>Period</i> _{Q2 2018}	1.1330 (1.1552)	1.6774 (1.2211)	−0.5956 (−0.4331)	0.3290 (0.2003)	1.6593 (1.1187)	1.8515 (0.8242)	−0.5050 (−0.5724)	0.6206 (0.6281)
<i>DTA Revaluations</i> _{<i>i</i>} × <i>Period</i> _{Q3 2018}	−1.3051 (−1.2781)	−0.2128 (−0.1400)	−2.8515** (−2.4669)	−0.7928 (−0.5494)	−0.4326 (−0.3649)	0.1735 (0.0775)	−2.9207*** (−2.8475)	−0.9823 (−0.8432)
<i>DTA Revaluations</i> _{<i>i</i>} × <i>Period</i> _{Q4 2018}	−1.1188 (−0.9956)	−1.7571 (−1.2447)	−1.4058 (−1.1437)	−0.2653 (−0.1864)	−0.8577 (−0.5414)	−2.0066 (−0.9209)	−1.2970 (−1.3159)	−0.6723 (−0.6077)
<i>DTA Revaluations</i> _{<i>i</i>} × <i>Period</i> ₂₀₁₉	−2.0338*** (−3.6876)	−1.9852 (−1.4710)	−1.2224 (−1.2913)	−0.6232 (−0.3951)	−1.7886** (−2.5658)	−2.2189 (−1.0348)	−1.6756** (−2.5724)	−1.2277 (−1.1157)
Additional controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State and quarter fixed effects	Yes		Yes		Yes		Yes	
Bank and quarter fixed effects		Yes		Yes		Yes		Yes
Observations	25,065	22,203	14,678	12,813	19,047	16,663	20,695	18,358
<i>R</i> ² (within)	0.0348	0.0385	0.0434	0.0463	0.0446	0.0364	0.0319	0.0445
<i>F</i>	39.80	19.91	29.61	17.59	41.60	15.89	27.67	20.65
Panel B: Tests on the differences in the estimation coefficients								
	(1) vs. (3)		(2) vs. (4)		(5) vs. (7)		(6) vs. (8)	
<i>DTA Revaluations</i> _{<i>i</i>} × <i>Period</i> _{Q4 2017}	−2.8327* [0.0656]		−2.5548 [0.7762]		−2.5621* [0.0896]		−2.4642 [0.7805]	

Notes. The table displays the estimation results from the following OLS panel for different subsamples:

$$\begin{aligned} DepVar_{i,t+1} = & \sum_{\tau=Q4\ 2017}^{2019} \beta_{\tau} Period_{\tau} * DTA\ Revaluations_i + DTA\ Revaluations_i + Income\ Taxes_{i,t} + Business\ Model\ Controls_{i,t} + Quarter\ FE_i \\ & + State/Bank\ FE_i + \varepsilon_{i,t}. \end{aligned}$$

The changes in (level of) banks’ total deposits_(t+1) serve as the dependent variable in the uneven (even) columns. Following Heese et al. (2022) in columns (1) through (4), we split the sample into geographic areas (US counties) with and without a local daily newspaper as in the UNC Hussman School of Journalism and Media database. Following Engelberg and Parsons (2011) in columns (5) through (8), we again split the results into geographic areas (the closest metropolitan area) with shares of local earnings coverage above and below the population median. Both variables approximate the differences in the information awareness costs. The quarter and state (bank) fixed effects are included in all uneven (even) specifications. Additional control variables as in Table 4 are included in the respective specifications. Wald tests on the differences in the estimation coefficients are reported in Panel B. The sample contains all U.S. commercial banks over the sample period from 2015 to 2019. Robust standard errors are clustered at the bank level. Robust *t* statistics [*p* values] are displayed in parentheses [brackets].

***, **, and *Significant at the 1% level, 5% level, and 10% level, respectively.

through (4)). We assume that local newspapers cover their local commercial banks that is supported by the anecdotal evidence provided in Appendix B.

Columns (1) and (2) of Table 5 show negative and statistically significant coefficients for our treatment variable $DTA\ Revaluations_i \times Period_{Q4\ 2017}$ in counties with local newspapers. In columns (3) and (4), there are also negative but lower and statistically nonsignificant coefficients in counties without local newspapers. In Panel B, we also see that the coefficients statistically differ between columns (1) and (3), whereas the large confidence intervals from column (4) yield statistically nonsignificant differences between columns (2) and (4). In untabulated tests, we show that this difference is driven by the banks from counties with local newspapers that have high circulations. Specifically, if we further split the subsample of banks with local newspapers based on the newspapers' circulations, we find a negative and significant difference between the high circulation group and the banks without newspaper exposure ($p = 0.039$), whereas the difference is smaller and nonsignificant for the group with below median circulation. That is, local newspapers can only reduce the IACs if a sufficient number of depositors read the newspapers. The results from columns (1) through (4) of Table 5 indicate that the depositors react to the banks' DTA revaluations by making larger withdrawals if their IACs are lower.

In columns (5) to (8) of Table 5, we use the nearest of the 19 cities for which Engelberg and Parsons (2011) have manually collected the propensity of the media to cover earnings news of local firms. The data by Engelberg and Parsons (2011) refer to the local earnings news coverage from the year 2007 and refers to all firms that makes this measure a noisy proxy for the coverage of earnings news in our sample of locally operating banks around the TCJA. We need to assume that the propensity of local earnings news positively correlates over time and applies to banks.

Columns (5) and (6) of Table 5 show similar results as in columns (1) and (2). There are negative and statistically significant coefficients for our treatment variable $DTA\ Revaluations_i \times Period_{Q4\ 2017}$ in areas with a probability of local coverage of earnings news above the median value (Engelberg and Parsons 2011). We find negative but statistically nonsignificant coefficients in columns (7) and (8) for geographic regions with a probability for local coverage of earnings news below the median value. Overall, we see the results from Table 5 as support for Hypothesis 2.

4.4. Variation in Depositors' Reactions to Their Costs of Integrating Information

Table 6 shows the reactions of different depositor groups to DTA revaluations due to the TCJA. Columns (1) and (2) present the effects for insured depositors,

columns (3) and (4) for uninsured depositors, and columns (5) and (6) present our benchmark results for the brokered deposits. There is a detailed discussion of brokered deposits and why they serve as a benchmark for highly sophisticated depositors in Online Appendix B. In Panel B of the table, we provide statistical tests to see whether the coefficients differ between the different types of depositors.

In columns (1) and (2), we find strong and significant withdrawals of insured deposits that are proportional to the amount of DTA revaluations from the TCJA. The $DTA\ Revaluations_i \times Period_{Q4\ 2017}$ has a negative and significant effect on the changes and on the level of deposits. Thereby, an average DTA revaluation due to the TCJA explains a reduction of 26.33% in insured deposits from the average change in insured deposits, which is also an economically significant change in banks' funding. Columns (3) and (4) show weaker negative coefficients for the uninsured depositors that are only statistically significant in column (3) when we use the change in deposits and state fixed effects. For the brokered deposits, we find the weakest negative or even a positive coefficient for our treatment $DTA\ Revaluations_i \times Period_{Q4\ 2017}$, and both remain nonsignificant far beyond conventional levels. The Wald test results in Panel B of Table 6 show that the coefficients from the change specifications in the uneven columns are also different in statistical terms between insured and brokered deposits and between uninsured and brokered deposits. The Wald tests for the specifications on the levels of deposits in the even columns yield only weak evidence. With a p value of 0.1508, the coefficients of our treatment effects from columns (2) and (6) differ in statistical terms above the conventional levels. Nevertheless, a lack of statistical differences can partially be attributed to the highly noisy coefficient in column (6). Overall, the results in Table 6 support Hypothesis 3 and indicate that the depositors' reaction to sophisticated but complex financial information is conditional on their level of processing costs: the higher the costs of integrating information, the more heuristically depositors react to arguably irrelevant information.

Table 7 presents the results on whether DTA revaluations due to the TCJA trigger larger withdrawals from depositors if they face higher IICs in the cross section. We approximate the cross-sectional differences in the IICs by looking at the share of the population with an occupational background that is relevant. We start by dividing the sample into banks located in areas with an above median share of people with a general business occupation (OCC code: 13-0000) from the *Occupational Employment and Wage Statistics 2017* by the Bureau of Labor Market Statistics. Second, we get even more specific and divide the sample by the population's share of occupations that explicitly shows a high expertise with

Table 6. Change in Different Deposit Types, Banks’ DTA Revaluations, and the TCJA

Panel A: Multivariate regression results						
Dependent variable	(1)	(2)	(3)	(4)	(5)	(6)
	<i>Insured Deposits_(t+1)</i>		<i>Uninsured Deposits_(t+1)</i>		<i>Brokered Deposits_(t+1)</i>	
	In changes	In levels	In changes	In levels	In changes	In levels
Treatment quarter						
<i>DTA Revaluations_i × Period_{Q4 2017}</i>	−1.3008*** (−3.0327)	−1.2358** (−2.0549)	−0.8830** (−1.9724)	−0.5407 (−0.8239)	0.1397 (0.6847)	−0.4328 (−1.0791)
Posttreatment effects						
<i>DTA Revaluations_i × Period_{Q1 2018}</i>	0.6442 (1.6099)	−0.4339 (−0.6468)	0.1079 (0.2234)	0.2580 (0.3740)	0.1576 (0.6265)	0.3715 (0.8009)
<i>DTA Revaluations_i × Period_{Q2 2018}</i>	0.2207 (0.4449)	−0.1938 (−0.2243)	0.4524 (0.9393)	1.1981* (1.7680)	0.2891 (1.3750)	0.4517 (0.9337)
<i>DTA Revaluations_i × Period_{Q3 2018}</i>	−0.5466 (−1.2630)	−0.5288 (−0.5305)	−1.0084** (−2.1065)	0.4466 (0.5737)	0.2181 (0.9250)	0.6193 (1.1714)
<i>DTA Revaluations_i × Period_{Q4 2018}</i>	−0.5296 (−1.1825)	−1.1955 (−1.2628)	−0.6873 (−1.4752)	0.2716 (0.3326)	−0.4744* (−1.7286)	0.2859 (0.5788)
<i>DTA Revaluations_i × Period₂₀₁₉</i>	−0.7586*** (−3.0256)	−1.1597 (−1.0737)	−0.6200** (−2.4798)	0.2671 (0.3452)	0.0145 (0.1147)	0.2699 (0.4593)
Additional controls	Yes	Yes	Yes	Yes	Yes	Yes
State and quarter fixed effects	Yes		Yes		Yes	
Bank and quarter fixed effects		Yes		Yes		Yes
Observations	39,743	35,137	39,743	35,137	39,743	35,137
R ² (within)	0.0470	0.117	0.0253	0.0829	0.0169	0.0220
F	44.91	41.34	26.11	31.09	13.27	7.238
Panel B: Tests on the differences in the estimation coefficients						
	(1) vs. (5)	(2) vs. (6)	(3) vs. (5)	(4) vs. (6)		
<i>DTA Revaluations_i × Period_{Q4 2017}</i>	−1.4405*** [0.0038]	−0.8030 [0.1508]	−1.0227** [0.0415]	−0.1079 [0.8524]		

Notes. The table displays the estimation results from the following OLS panel model:

$$DepVar_{i,t+1} = \sum_{\tau=Q4\ 2017}^{2019} \beta_{\tau} Period_{\tau} * DTA\ Revaluations_i + DTA\ Revaluations_i + Income\ Taxes_{i,t} + Business\ Model\ Controls_{i,t} + Quarter\ FE_t + State/Bank\ FE_i + \varepsilon_{i,t}.$$

The changes in (level of) insured retail deposits_(t+1) (columns (1) and (2)), uninsured deposits_(t+1) (columns (3) and (4)), and brokered retail deposits_(t+1) (columns (5) and (6)) serve as the dependent variables. The quarter and state [bank] fixed effects are included in all uneven [even] specifications. Additional control variables as in Table 4 are included in the respective specifications. Wald tests on the differences in the estimation coefficients are reported in Panel B of the table. The sample includes all U.S. commercial banks over the period from 2015 to 2019. Robust standard errors are clustered at the bank level. Robust *t* statistics [*p* values] are displayed in parentheses [brackets].

***, **, and *Significant at the 1% level, 5% level, and 10% level, respectively.

respect to financial accounting. In Panel B, we provide statistical tests to see whether the coefficients differ between the subsamples.

From columns (1) through (4) of Table 7, we see that the depositors’ reactions to the DTA revaluations due to the TCJA strongly differ because of the IICs. We see that banks experience less withdrawals in the areas where the population’s general business expertise measured by their occupation is higher (columns (1) and (2)), whereas we find highly significant and large withdrawals in areas with a lower business expertise (columns (3) and (4)). Also, when looking only at occupations with the strongest expertise in financial accounting in columns (5) through (8), the results show large differences in the

depositors’ reactions to DTA revaluations that are conditional on their IICs. Overall, the evidence from Tables 6 and 7 provides strong support for Hypothesis 3 and shows that depositors react more strongly to banks’ DTA revaluations if their IICs are higher.

4.5. Additional Economic Consequences of the DTA Revaluations

The results in Table 8 show how the banks react to the previously described change in deposit levels due to the DTA revaluations under the TCJA. Therefore, we analyze whether the reactions from depositors who (most likely rationally) underinvest in integrating information have additional economic consequences. In column (1),

Table 7. Depositors' Reactions and Information Integration Costs

Dependent variable: <i>Total deposits</i> _{<i>t</i>+1}	Panel A							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Share of business occupations				Share of financial accounting occupations			
	Above median		Below median		Above median		Below median	
	In changes	In levels	In changes	In levels	In changes	In levels	In changes	In levels
Treatment quarter								
<i>DTA Revaluations</i> _{<i>i</i>} × <i>Period</i> _{Q4 2017}	−1.8975** (−2.1209)	−0.3752 (−0.3680)	−5.3610*** (−2.9445)	−4.3707*** (−2.6851)	−1.6562** (−2.1287)	−0.3302 (−0.2967)	−4.5592*** (−3.4180)	−3.6879*** (−2.9983)
Posttreatment effects								
<i>DTA Revaluations</i> _{<i>i</i>} × <i>Period</i> _{Q1 2018}	1.5728 (1.6096)	1.8936* (1.8893)	−1.8545 (−1.1986)	−1.3413 (−0.5690)	0.7682 (0.8023)	2.0098 (1.5978)	0.9367 (0.6336)	−0.7412 (−0.5842)
<i>DTA Revaluations</i> _{<i>i</i>} × <i>Period</i> _{Q2 2018}	−0.6092 (−0.8808)	1.2163 (1.6284)	5.7243*** (2.6100)	6.1809** (2.2631)	−0.6711 (−0.8044)	1.4634 (1.4661)	3.4078** (2.4510)	3.3602** (2.1519)
<i>DTA Revaluations</i> _{<i>i</i>} × <i>Period</i> _{Q3 2018}	−0.6161 (−0.6392)	1.0601 (1.1511)	−0.0737 (−0.0514)	4.6786* (1.9553)	−0.1000 (−0.0950)	1.5522 (1.5482)	−1.2610 (−0.9754)	1.7356 (0.9961)
<i>DTA Revaluations</i> _{<i>i</i>} × <i>Period</i> _{Q4 2018}	−0.4647 (−0.4924)	1.0064 (1.1418)	−4.6841** (−2.3903)	−0.5276 (−0.1847)	0.0738 (0.0849)	1.8472** (2.2613)	−3.4584** (−2.3397)	−1.8471 (−1.1458)
<i>DTA Revaluations</i> _{<i>i</i>} × <i>Period</i> ₂₀₁₉	−1.1933** (−2.1928)	0.5365 (0.5671)	−2.5723** (−2.2876)	−0.8682 (−0.3186)	−0.9674 (−1.6332)	1.2935 (1.3960)	−1.9623** (−2.4442)	−1.4132 (−0.9541)
Additional controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State and quarter fixed effects	Yes		Yes		Yes		Yes	
Bank and quarter fixed effects		Yes		Yes		Yes		Yes
Observations	9,611	9,460	9,549	9,425	9,523	9,395	9,637	9,490
<i>R</i> ² (within)	0.0538	0.0420	0.0510	0.0409	0.0577	0.0612	0.0559	0.0563
<i>F</i>	9.865	6.274	14.77	13.35	9.501	6.551	12.58	11.21
Panel B: Tests on the differences in the estimation coefficients								
	(1) vs. (3)		(2) vs. (4)		(5) vs. (7)		(6) vs. (8)	
<i>DTA Revaluations</i> _{<i>i</i>} × <i>Period</i> _{Q4 2017}	3.0801 [0.125]		3.9956** [0.037]		2.732* [0.073]		3.3577** [0.043]	

Notes. The table displays the estimation results from the following OLS panel model for different subsamples:

$$\begin{aligned}
 \text{DepVar}_{i,t+1} = & \sum_{\tau=Q4\ 2017}^{2019} \beta_{\tau} \text{Period}_{\tau} * \text{DTA Revaluations}_i + \text{DTA Revaluations}_i + \text{Income Taxes}_{i,t} + \text{Business Model Controls}_{i,t} + \text{Quarter FE}_t \\
 & + \text{State/Bank FE}_i + \varepsilon_{i,t}.
 \end{aligned}$$

The changes in (level of) banks' total deposits_{*t*+1} serve as the dependent variable in the uneven (even) columns. The table splits the sample by the median share of local professionals with general business occupations (columns (1) through (4)) and more specific financial accounting (columns (5) through (8)). All variables approximate the differences in the costs of integrating information. Differences between areas with higher compared with lower IICs are tested in Panel B of the table. The quarter and state (bank) fixed effects are included in all uneven (even) specifications. Additional control variables as in Table 4 are included in the respective specifications. The sample contains all U.S. commercial banks over the period from 2015 to 2019. Robust standard errors are clustered at the bank level. Robust *t* statistics (*p* values) are displayed in parentheses (brackets).

***, **, and *Significant at the 1% level, 5% level, and 10% level, respectively.

we first investigate whether banks with DTA revaluations offer higher deposit rates compared with banks without DTA revaluations to compensate for the deposit withdrawals in the subsequent periods. In column (2), we investigate whether the banks that are affected by the DTA revaluations also react by subsequently granting fewer loans.¹⁵

Column (1) of Table 8 shows a positive and statistically nonsignificant coefficient for *DTA Revaluations*_{*i*} × *Period*_{Q4 2017} on banks' deposit rates (*p* = 0.126). The

coefficient remains positive in all subsequent quarters and turns statistically significant in periods Q1 2018, Q4 2018, and 2019. The results of these posttreatment effects indicate that the withdrawals of deposits after the DTA revaluations affected the banks and led them to offer higher deposit rates to their depositors after the TCJA and that these differences persisted for multiple quarters. Those higher rates arguably were to reattract lost depositors or to attract new depositors. Column (2) shows that *DTA Revaluations*_{*i*} × *Period*_{Q4}

Table 8. Other Economic Consequences and Banks’ Reaction to the Deposit Withdrawals

Dependent variable	(1) <i>Deposit Rates</i> _(t+1)	(2) <i>Changes in Total Loan</i> _(t+1)
Treatment quarter		
<i>DTA Revaluations</i> _{<i>i</i>} × <i>Period</i> _{Q4 2017}	0.0459 (1.5316)	−2.2673* (−1.7393)
Posttreatment effects		
<i>DTA Revaluations</i> _{<i>i</i>} × <i>Period</i> _{Q1 2018}	0.0689* (1.7989)	−2.3172* (−1.6766)
<i>DTA Revaluations</i> _{<i>i</i>} × <i>Period</i> _{Q2 2018}	0.0653 (1.4633)	−2.3957* (−1.6630)
<i>DTA Revaluations</i> _{<i>i</i>} × <i>Period</i> _{Q3 2018}	0.0659 (1.3003)	−0.7323 (−0.5671)
<i>DTA Revaluations</i> _{<i>i</i>} × <i>Period</i> _{Q4 2018}	0.0977* (1.8718)	−1.2141 (−0.9162)
<i>DTA Revaluations</i> _{<i>i</i>} × <i>Period</i> ₂₀₁₉	0.1008* (1.9401)	−3.5917** (−2.2891)
Additional controls from Chen et al. (2022)	Yes	Yes
Bank and quarter fixed effects	Yes	Yes
Observations	34,858	28,254
<i>R</i> ² (within)	0.0940	0.0335
<i>F</i>	18.79	8.987

Notes. The table displays the estimation results from the following OLS panel model:

$$DepVar_{i,t+1} = \sum_{\tau=Q4\ 2017}^{2019} \beta_{\tau} Period_{\tau} * DTA\ Revaluations_i + DTA\ Revaluations_i + Controls_{i,t} + Quarter\ FE_t + Bank\ FE_i + \varepsilon_{i,t}.$$

The deposit rate_(t+1) and the changes in banks’ loans_(t+1) serve as the dependent variables. The quarter and bank fixed effects are included in all specifications. Additional control variables are banks’ shares of wholesale funding, shares of unused commitments, shares of retail loans, the amount of leverage, return on assets as in Chen et al. (2022) are included in all specifications. Further, the use of period fixed effects replaces the macroeconomic control variables in the models by Chen et al. (2022). The sample contains all US commercial banks over the period from 2015 to 2019. Robust standard errors are clustered at the bank level. Robust *t* statistics are displayed in brackets.

***, **, and *Significant at the 1% level, 5% level, and 10% level, respectively.

2017 has a negative and significant treatment effect on the change in total loans and *DTA Revaluation*_{*s*_{*i*}} × *Period*_{Q1 2018} and *DTA Revaluations*_{*s*_{*i*}} × *Period*_{Q2 2018} also has negative and significant posttreatment effects. Overall, the results show adverse additional economic consequences from the hasty reactions of underinformed depositors that were due to the DTA revaluations following the TCJA.

5. Alternative Mechanisms and Other Robustness Tests

If banks anticipate the effects of the TCJA and the degrees of anticipation correlate with unobserved characteristics, our regression coefficient might partially reflect these unobserved characteristics. To alleviate concerns that these anticipation effects drive our results, we capture the influence of the DTA revaluations due to the TCJA with a dummy variable to measure the binary difference between banks that reported net DTAs in the first quarter of 2016 and those that did not report net DTAs. In the first quarter of 2016, the outcome of the presidential election was unexpected and, consequently, the change in the tax code was unexpected as well. Thus, the interaction term represents a classic difference-in-

differences approach and captures the average difference in the deposit growth between the banks that were more likely to be treated (i.e., they already report net DTAs in Q1 2016) and the banks that were less likely to be affected by the DTA revaluations due to the TCJA. Unreported evidence indicates that our inferences remain unaffected using this alternative specification, which is less affected by concerns regarding endogeneity. Our results also hold if we transform our treatment variable into deciles or quantiles, which alleviates concerns about alternative functional forms (see Online Appendix C.2).

The valuation allowance accounts in net DTA might potentially distort our measure of changes in net DTAs if banks face long sequences of losses. In line with this argument, Dhaliwal et al. (2013) show that under certain conditions, the classification of the valuation allowance that accounts for net DTAs contains private and forward-looking information about banks’ persistent accounting losses. Because our treatment uses the stock of DTAs at the end of the third quarter and does not rely on banks’ self-reported changes in net DTAs, our treatment variable should be unaffected by the valuation allowance accounts as discussed in Dhaliwal et al. (2013). We also empirically address this concern by excluding all bank quarters with losses from our sample

in columns (1) and (2) of Panel A in Online Appendix C.3 and even all banks that ever report a loss in a quarter during our sample period in columns (3) and (4) of Online Appendix C.3. Because banks experienced a stable economic environment and good business outlooks in the years 2015 through 2019, there are only a few observations. The results from columns (1) through (4) in Panel A of Online Appendix C.3 show that the results and inferences are not materially affected.

Simultaneous shocks from other events, such as the 2018 Economic Growth, Regulatory Relief, and Consumer Protection Act (henceforth: EGRRCPA), coincide with our treatment and, consequently, are a major concern for any empirical identification strategy. Because we rely on a fuzzy difference-in-differences approach with different treatment intensities for different amounts of DTA revaluations due to the TCJA, our approach is less exposed to simultaneous shocks than a standard difference-in-differences approach with binary treatments. Any simultaneous event only affects our direct treatment effect if the event's effect on deposits is proportional to the bank's DTAs revaluations due to the TCJA. Because the EGRRCPA does not change banks' DTAs revaluations, our empirical results cannot be because of it.

The 2017 incorporation of the new Basel III rules into U.S. law happens slightly before our treatment at the end of 2017, but Basel III has gradually implemented changes over the sample period that specifically pertained to DTAs and banks' regulatory capital. Following the implementation of Basel III, banks had to also deduct net DTAs that would be realized within the coming 12 months and that exceeded 10% of their Tier 1 capital from their regulatory capital. Further, the net DTAs realized later are deducted anyhow throughout our sample period. First, banks rarely report net DTAs realized within the coming 12 months that will exceed 10% of their Tier 1 capital; consequently, the implementation of the new Basel III barely affected any banks (Klingler 2017). Second, we explicitly control for the equity ratio in all specifications but, in additional tests, we also control for the Tier 1 capital ratio. This variable should already capture most of the effects. All our empirical inferences remain unchanged if we explicitly control for banks' Tier 1 capital ratios. Third, in additional tests, we exclude all banks that report a ratio of net DTAs to Tier 1 capital of more than 10% in any quarter. The results remain qualitatively unchanged.

Poterba et al. (2011) argues that firms with DTAs have incentives to shift income to the high tax period to use deferred tax benefits at the current high tax rate that can create an indirect effect on the banks' risk taking. However, in our setting, this is arguably unlikely because the TCJA was passed after a very short legislative process in November 2017. After the TCJA was passed, banks had only limited time to adjust their real

activities (e.g., accelerating revenues by engaging in high-risk activities) before the end of the fiscal year. Thus, income shifting via accrual earnings management and similar activities (Gaertner et al. 2020) were more likely. Consequently, a potentially accelerated recognition of income would not affect banks' long-term risk-taking but would only affect the earnings signal. The earnings signal would be affected in the opposite direction of the DTA revaluation and thus would work against finding an effect from DTA revaluations. The changes in business models in anticipation of the TCJA would happen gradually over the previous quarters and, consequently, are not limited to the TCJA quarter.

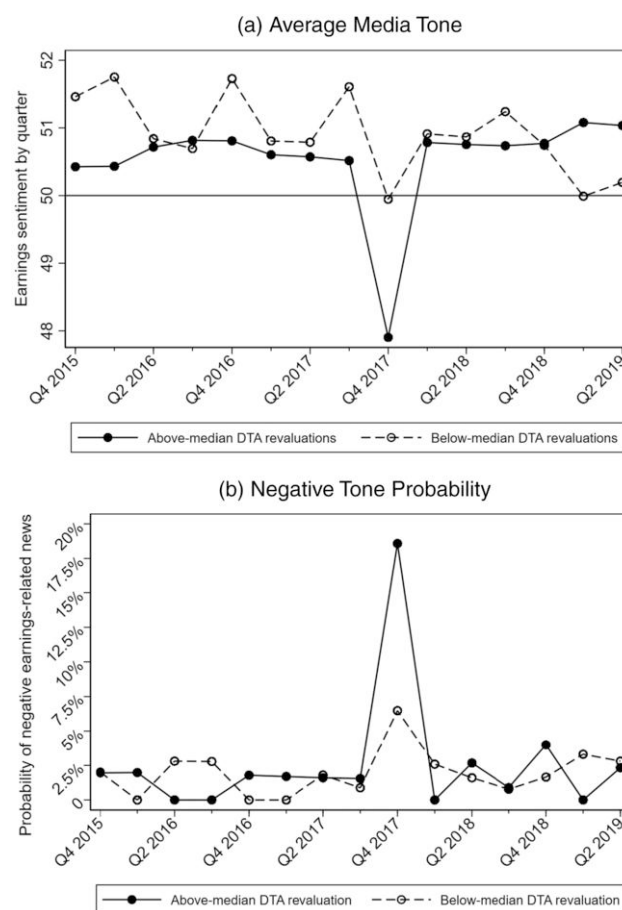
Our treatment captures the DTA revaluations due to the TCJA based on the DTA stock at the end of the third quarter of 2017. One other concern goes back to alternative influences of the TCJA on banks' earnings management at the end of 2017 that might simultaneously affect banks' DTAs. In line with this argument, Gaertner et al. (2020) show that firms made large pension contributions at the end of 2017 to avoid DTA write-downs. To avoid any contamination of our results from earnings management at the end of 2017, we reestimate our results on the treatment effect by calculating the DTA revaluations by using the DTA stock from the first to the end of the second quarter of 2017 in alternative specifications. Earnings management due to the TCJA should not affect the DTA stock from the beginning of 2017. In the additional tests, we find that the results and inferences are not materially affected, although the coefficients for the treatment effect become slightly smaller and weaker in terms of statistical significance. The lower coefficients and the weaker inferences point to more noise in these alternative specifications. We see this as evidence that other types of earnings management due to the TCJA do not provide an alternative explanation for our results. In further tests, we find that our empirical results remain unchanged if we limit our sample to c-corporation banks to obtain a more homogenous sample.

A different concern refers to the origin of the DTAs. The DTAs come from temporary differences in banks' financial and tax accountings but also from net operating losses. The firms with large amounts of DTAs from net operating losses might systematically differ from the typical firm (Hanna et al. 2019) that would reduce the external validity of our results. We already control for net operating income to alleviate concerns that our results are driven by these firms with large operating losses. We additionally address this concern empirically by dividing our treatment into the DTAs that come from temporary differences between the financial and the tax accountings and the DTAs that come from net operating losses. We display the results in columns (5) and (6) of Online Appendix C.3 in Panel A. We see

that the treatment effects are negative for both subcomponents of DTAs. Nevertheless, the effect from DTAs from net operating losses is less negative and statistically not significant while the DTAs from temporary differences show a more negative and statistically significant effect on banks' total deposits. The results provide empirical support that our results can be attributed to banks with DTAs from temporary differences.

Another concern goes back to the underlying channel behind our results. We argue that depositors do not fully process the information from the line items in the bank's financial statement. Nevertheless, local journalists might already have failed to incorporate deferred tax changes in their newspaper articles and the depositors exclusively react to the local news coverage. The earnings news for our sample of commercial banks gets mostly covered by robo journalistic articles (Blankespoor et al. 2018) like those in Appendix B. These technologies may disregard a lot of relevant information that human reporters would likely catch (Guest 2021). Similarly, the literature shows that financial analysts also got misled by the DTA effect on firms' net income due to the TCJA when doing their forecasts (Chen and Koester 2021). Consequently, the depositors could also react to the already biased reports of analysts, either directly or from other news outlets. We cannot fully rule out these alternative channels, but they yield similar empirical predictions. Especially, both channels result in depositors' reactions that disregard the line items in the income statement. Figure 5 shows the average media tone and the probability of receiving negative media coverage on earnings news around the TCJA for the commercial banks that belong to publicly listed BHCs. The results indicate that the earnings news became more negative in the fourth quarter of 2017 and returned to the pre-TCJA level afterward. Moreover, the tone of the media coverage becomes more negative for banks with larger DTA revaluations due to the TCJA. Second, the depositors' reactions are stronger, not weaker, for banks in areas with stronger earnings coverage and more local newspapers that is consistent with bank depositors not understanding the underlying economic reality of the DTA revaluations from the local media, for example, because the DTA revaluations were too complex. These results are consistent with at least some newspaper articles not highlighting the positive economic consequences of the TCJA for the banks but stressing the negative short-term net income consequences that resulted in a more negative sentiment. Concerning the second channel, the vast majority of our commercial banks receive no direct coverage because analysts typically do not cover individual commercial banks but only listed BHCs. Therefore, we perceive these alternative channels as less likely but still plausible.

Figure 5. Media Tone on Bank Earnings Announcements Around the TCJA



Notes. (a) Average tone of the earnings-related media coverage. It is the average media event sentiment score (ESS) for events that refer to bank earnings. (b) Probability of receiving media coverage with a negative tone on an earnings-related event. It is the probability of having at least one earnings-related media article in a quarter with an event sentiment score (ESS) that is below 50. The x axis refers to the period of the earnings information (e.g., Q4 2017) that is picked up in the media once the earnings gets released (e.g., Q1 2018).

The heterogeneity in depositors' reactions of withdrawing and returning their deposits is important and interesting to understand. In contrast to the studies by Iyer et al. (2016), Iyer and Puri (2012), and Martin et al. (2018) that investigate depositors' reactions based on their account data at one or two specific banks, we do not have access to deposit account data for our sample banks. Consequently, we can compare banks only based on the aggregated bank data. Iyer et al. (2016), among many others, have shown the differences in the stickiness of deposits, for example, older deposit accounts and depositors with multiple bank products are less volatile. To see whether the effects for our empirical tests are consistent with the differences in the stickiness of depositors (Iyer et al. 2016), we split our sample by the standard deviation in banks' share of deposits. Thereby,

we compare banks with a higher-than-median standard deviation in deposits, that is, with more volatile deposits, to the banks with a lower-than-median standard deviation in deposits, that is, a more stable deposit basis. The results in Panel B of Online Appendix C.3 indicate that the effect of DTA revaluations due to the TCJA on the depositors' reaction is clearer for the subsample of banks with less sticky deposits. Nevertheless, the coefficients for the treatment effects are not statistically different at conventional levels. Overall, the evidence appears at least consistent with the effects documented by Iyer et al. (2016).

6. Conclusion

The depositors react to accounting information (Chen et al. 2022) and their reactions potentially influence how bank transparency affects market monitoring and financial stability (Bushman 2014, Acharya and Ryan 2016). Yet, we have little understanding of how depositors process accounting information despite the other studies identify depositors' responses to banks' financial reporting (Chen et al. 2022). In this study, we use the DTA revaluations from the Tax Cuts and Jobs Act as a showcase of how the costs of processing information affect the extent to which accounting signals get incorporated into depositors' reactions. These costs can help to mute or to increase how bank depositors react to complex information.

We provide strong evidence in a difference-in-differences approach that depositors with higher costs for financial integration and lower costs for information awareness react to complex signals from financial accounting that do not contain any incrementally relevant information about risk. Our results are consistent with either depositors being rationally inattentive to the underlying information content of these signals or having potential behavioral explanations (Banerjee et al. 2020).

One question relates to the generalizability of the results. First, there are other accounting signals that are costly for depositors to integrate. Because depositors barely acknowledge additional explanations, our setting most likely prevents banks from increasing the clarity with additional explanations, although this might be possible in other settings. It is also an open empirical question as to whether complex accounting choices with a strategic component, for example tax planning (Gaertner et al. 2020), have similar consequences on the depositor reactions to financial information. The consequences of these choices are more difficult to investigate empirically because accounting information cannot be separated from management's decisions (Acharya and

Ryan 2016, Blankespoor et al. 2020). The TCJA permits a clear identification in a difference-in-differences design that is typically unavailable in these other situations.

Overall, we show in this study that the costs of processing information can shift the balance between the disclosures by banks that promote or impair the market monitoring by depositors. We speak to the debate on the general presentation of financial statements and disclosures (IFRS Foundation 2019) to underscore the importance of their attempt to provide items and subtotals that are easier to follow.

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Appendix A. Example of the DTA Revaluations in the Statement of Profit and Losses

Appendix A provides a screenshot from the quarterly earnings report of one bank from the sample (Figure A.1). The bank is the "Capital One Financial Corporation" and the information comes from the financial supplement from the fourth quarter of 2017 obtained from the bank's website (<https://investor.capitalone.com/>). The additional boxes highlight the items that potentially guide the depositors, that is, the net interest income, the pre-tax total income, and the bottom-line net income (loss) after taxes. The additional arrow shows the line item that captures the DTA revaluations due to the TCJA for which the depositors need to adjust their assessment of the bank's economic situation because these DTA revaluations are not risk-relevant to the bank. If depositors correctly process the line-item information or are guided by the items of net interest income or income from continuing operations before taxes, we expect no reactions from them to the DTA revaluations. Nevertheless, the example in Appendix A highlights that the financial supplement contains a lot of information that is not easy to process, and the bottom-line net income provides a tempting, yet potentially misleading, piece of information.

Figure A.1. (Color online) Example of the Effects of DTA Revaluations in the Income Statement



	2017	2017	2017
	Q4	Q3	Q2
(Dollars in millions, except per share data and as noted)			
Income Statement			
Net interest income	\$ 5,813	\$ 5,700	\$ 5,473
Non-interest income	1,200	1,285	1,231
Total net revenue ⁽¹⁾	7,013	6,985	6,704
Provision for credit losses	1,926	1,833	1,800
Non-interest expense:			
Marketing	460	379	435
Operating expenses	3,319	3,188	2,979
Total non-interest expense	3,779	3,567	3,414
Income from continuing operations before income taxes	1,308	1,585	1,490
Income tax provision	2,170	448	
Income (loss) from continuing operations, net of tax	(862)	1,137	1,047
Income (loss) from discontinued operations, net of tax ⁽²⁾	(109)	(30)	(11)
Net income (loss)	(971)	1,107	1,036

Source. Capitol One website (<https://investor.capitalone.com/>).

Appendix B. Example of Medias’ Earnings News Coverage

Appendix B shows examples of the media coverage on banks’ earnings information around the TCJA (Figures B.1 and B.2). In black, we display all relevant coverage of earnings information. In gray, we display the remaining media coverage that, according to our judgement, does not refer to the bank earnings or the TCJA.

Example B.1. Example 1 of medias’ earnings news coverage

Title:	Earnings: Union Bankshares and Capital One Financial
Media Outlet:	Richmond Times Dispatch (Virginia)
Date:	January 24, 2018

Figure B.1. 1st Example of Medias’ Earnings News Coverage

Richmond-based firm: Union Bankshares

Union Bankshares Corp., the Richmond-based parent company of Union Bank & Trust, reported fourth-quarter net income of \$15.2 million, or 35 cents per share, compared with net income of \$20.8 million, or 48 cents per share, in the year-earlier quarter. For the year that ended Dec. 31, net income was \$72.9 million, or \$1.67 per share, compared with net income of \$77.5 million, or \$1.77 per share, for the previous year. The fourth quarter and full year results do not include the financial results of Xenith Bankshares Inc., which the company acquired on Jan. 1. Union Bank & Trust has 150 banking offices in Virginia, Maryland and North Carolina.

Major local employer: Capital One Financial

Capital One Financial Corp. reported a fourth-quarter loss of \$971 million because of credit losses and recent changes to the U.S. tax law. The McLean-based company, which is the Richmond area's largest private employer, said it took a hit of \$1.77 billion related to the recent tax code changes. The company reported a net loss of \$971 million, or a loss of \$2.17 per share, compared with a profit of \$791 million, or \$1.45 per share, in the fourth quarter of 2016. Earnings, adjusted for one-time gains and costs, were \$1.66 per share in the fourth quarter of 2017.

The results missed Wall Street expectations. The average estimate of 10 analysts surveyed by Zacks Investment Research was for earnings of \$1.85 per share. The credit card issuer and bank posted revenue of \$7.8 billion in the period. Its adjusted revenue was \$7.01 billion, which also did not meet Wall Street forecasts. Eight analysts surveyed by Zacks expected \$7.12 billion. For the year, the company reported a profit of \$1.98 billion, or \$3.49 per share. Revenue was \$27.24 billion.

Example B.2. Example 2 of medias’ earnings news coverage

Title: **Banner Corp. reports 4Q loss, cites tax expense**
Media Outlet: *Spokesman Review* (Spokane, Washington)
Date: January 26, 2018

Figure B.2. 2nd Example of Medias’ Earnings News Coverage

Banner Bank's parent company reported a \$13.5 million loss during the fourth quarter, which officials attributed to a one-time tax expense.

Banner Corp. took a write down of deferred tax assets following the passage of the federal Tax Cuts and Jobs Act in December, which resulted in additional tax expense of \$42.6 million. Earnings were also affected by the company's sale of its seven Utah branches and related operations, which generated a one-time gain of \$12.2 million.

The Walla Walla-based company reported a net loss of 41 cents per share during the fourth quarter, compared to 2016 fourth-quarter income of \$22.8 million, or 69 cents per share.

For 2017, Banner's net income was \$60.8 million, or \$1.84 per share, compared to 2016 income of \$85.4 million, or \$2.54 per share.

The company operates 178 branches of Banner Bank and Islanders Bank.

Appendix C. Descriptions and Sources of Variables

Variable	Description	Source and item
<i>(Change in) the Level of Total Deposits_{i,t}</i>	(Change in) the level of total deposits scaled by the banks’ total assets.	FFIEC (rcon 2200)
<i>(Change in) the Level of Insured Deposits_{i,t}</i>	(Change in) the level of insured deposits for all deposits up to 250,000 USD per account that are insured by the FDIC scaled by the banks’ total assets.	FFIEC (rcon f049 and rcon f045)
<i>(Change in) the Level of Uninsured Deposits_{i,t}</i>	(Change in) the level of uninsured deposits for all deposits that exceed 250,000 USD per deposit account scaled by the banks’ total assets.	FFIEC (rcon f051 and rcon f047)
<i>(Change in) the Level of Brokered Deposits_{i,t}</i>	(Change in) the level of brokered deposits for the changes in those deposits that are managed by a professional deposit broker scaled by the banks’ total assets.	FFIEC (rcon 2365)
Period _τ	Different indicator variables that equals one for each period at and after the TCJA where τ is either Q4 2017, Q1 2018, Q2 2018, Q3 2018, Q4 2018, or 2019. We refrain from using indicator variables for each quarter of 2019 to enhance the readability of our tables. In the regression outputs, the base terms of the indicator variables are captured by the quarter fixed effects and not reported separately.	Own calculation based on the FFIEC filing period.
<i>DTA Revaluations_i</i>	This variable captures the treatment’s intensity, that is, the amount of DTA revaluations that a bank experienced from the Tax Cuts and Jobs Act. Therefore, we use the net DTAs from September (Q3) of 2017 multiplied by the change in the tax rate if the banks reported net DTAs in the third quarter of 2017 and zero otherwise. Thereby, we rule out any competing influences from newly generated DTAs due to the TCJA as described by Gaertner et al. (2020). If banks report net DTL in Q3 2017, we set the value to zero to avoid contaminating the effect with positive earnings shocks. In additional tests, we show that our results are not robust with respect to this measurement choice.	Own calculation based on the FFIEC items (rcon 2148; 3049)

Appendix C. (Continued)

Variable	Description	Source and item
<i>Income Taxes_{i,t}</i>	This variable captures the applicable income taxes (annualized) reported in the specific quarter scaled by beginning total assets.	FFIEC (riad 4302)
<i>(Change in) the Level of Net DTAs_{i,t}</i>	(Change in) the level of DTAs minus DTLs scaled by beginning total assets	FFIEC (rcon 2148; 3049)
Additional controls		
<i>Operating Income before Taxes_{i,t}</i>	Operating income (loss) before applicable income taxes and discontinued (annualized) scaled by beginning total assets	FFIEC (riad 4301)
<i>(Change in) Size_{i,t}</i>	(Change in) the natural logarithm of banks' total assets	FFIEC (rcon 2170)
<i>(Change in) the Equity Ratio_{i,t}</i>	(Change in) the banks' equity to total assets.	FFIEC (rcon 3210)
<i>(Change in) Loan-to-Asset Ratio_{i,t}</i>	(Change in) the variable that indicates the ratio of loans and leases (both, held for sales and held for investment) net of loan loss allowances on these loans scaled by beginning total assets.	FFIEC (rcon b528; 5369; 3123)
<i>(Change in) Nonperforming Loans_{i,t}</i>	(Change in) the nonperforming loans scaled by current gross loans following Brushwood et al. (2020)	FFIEC (rcon 3495; 5400; c229; c230; 3501; f182; f183; b836; 1608; b577; k215; k218; 5461; 1228)
<i>Loan Loss Provisions_{i,t}</i>	Provisions for credit losses scaled by current gross loans.	FFIEC (riad 4230 to rcon 2122)
<i>Loan Charge-Offs_{i,t}</i>	The charge-offs net of recoveries on loans and leases scaled by current loan loss allowances,	FFIEC (riad 4635 to rcon 3123)
Other dependent variables and controls		
<i>Deposit Rates_{i,(t+1)}</i>	This variable measures the interest expenses on all deposits (including transaction accounts, savings deposits, and time deposits) during a rolling window of four quarters per year scaled by total average deposits.	FFIEC (riad 0093; hk03; hk04; a518; 4508)
<i>Non-Interest Income_{i,t}</i>	The current share of noninterest income that influences the future level of deposit rates and loans according to the literature.	FFIEC (riad 4074)
<i>(Changes in) Total Loan_{i,(t+1)}</i>	This variable indicates the (changes in) loans and leases (both, held for sales and held for investment) net of loan loss allowances on these loans scaled by beginning total assets.	FFIEC (rcon b528; 5369; 3123)
Additional variables		
<i>Deposit Stickiness_i [Less / More Sticky]</i>	Less [More] Sticky are binary variables that equal one if the bank shows a standard deviation in total deposits before the TCJA (Q1 2016 and Q3 2017) that is below [above] the population's median value.	FFIEC (rcon 2200)
<i>Earnings News Coverage_i</i>	The natural logarithm of one plus the share of local earnings coverage in percentage points. The earnings coverage probability is a geographical proxy and refers to the share of local firms with coverage of earnings news for the year 2007. The data represent the largest 19 U.S. cities.	Engelberg and Parsons (2011)
<i>Local Newspapers_i</i>	Dummy variable that equals one in regions with at least one locally focused, daily newspaper, and zero otherwise.	the UNC's Project OASIS, see also Heese et al. (2022).
<i>Share of Business Occupations_i</i>	Metropolitan Statistical Area that shows the share of local professionals with general business occupations (Occupation Code: 13-0000) that is above / below the median value in 2017.	Bureau of Labor Statistics; the Occupational Employment and Wage Statistics (Occupation Code)
<i>Share of Financial Accounting Occupations_i</i>	Metropolitan Statistical Area that shows the share of local professionals with financial accounting related occupations (accountants and auditors, actuaries, bookkeeping, accounting, and auditing clerks, economists, financial and investment analysts, financial risk specialists, and financial specialists, tax examiners and collectors, and revenue agents, and tax preparers) that is above/below the median value in 2017.	Bureau of Labor Statistics; the Occupational Employment and Wage Statistics (OCC Title)

Endnotes

¹ It is not necessary that the depositors fully understand the cash-flow consequences of these DTA revaluations. It is sufficient if they either understand that the DTA revaluations are a one-time effect or, alternatively, that they do not materially affect banks' ability to repay the deposits.

² DTA revaluations are also unaffected by most earnings management or signaling incentives (see Herbohn et al. (2010) for the strategic use of DTAs). We explicitly deal with some earnings management incentives prior to the TCJA in additional analyses.

³ We limit our investigation to domestic banks because these banks are unaffected by changes in the tax rules for foreign operations and subsidiaries. Nevertheless, our inferences hold if we add the domestic operations of multinational banks.

⁴ Our measure of information awareness costs also captures parts of the information acquisition costs because local newspapers also distribute the earnings information. Nevertheless, information acquisition costs also include other important dimensions that refer to costs in obtaining the information, for example, through databases (Blankespoor et al. 2019). Throughout the paper, we refer to this measure as a measure of information awareness costs but openly acknowledge that it also captures aspects of information acquisition costs.

⁵ The treatment of DTAs is similar in most countries. An exception is Japan during the Asian crisis that had severe consequences for Japanese banks as documented by Skinner (2008).

⁶ In additional tests, we also replicate the results by Wagner et al. (2018) with our sample and show that the market only reacted to the cash-flow relevant tax effects around Donald Trump's election in 2016, that is, when the information entered the market for the first time.

⁷ Iyer and Puri (2012) show that depositors ran from a fundamentally sound bank when a nearby bank failed and that the deposit insurance was only partly successful at limiting the outflow of insured deposits. Baer and Brewer (1986) and Ellis and Flannery (1992) provide earlier evidence on the market discipline by uninsured depositors; and Baer and Brewer (1986), Cook and Spellman (1994), Maechler and McDill (2006), and Park (1995) provide similar evidence for insured depositors. Bischof et al. (2023) show that insured depositors exert market discipline to improve financial reporting quality of privately held banks.

⁸ The 2019 survey found an average of 5.3 accounts at banks and other financial institutions per consumer (Mercator Advisory Group 2019).

⁹ In line with this argument, venture capitalists' Founders Fund reportedly urged its start-ups to pull their deposits early from the Silicon Valley bank (SVB) in March 2023 because there was "no downside" to pulling the money away from SVB. Bloomberg News (March 11, 2023).

¹⁰ Bank depositors might also react to the TCJA in of itself. Hutchens et al. (2024) find lower pay satisfaction of employees around the TCJA that can result in small depositors pulling out funds and consuming more to feel better. Because depositors' direct reactions to the TCJA are not conditional on banks' DTA revaluations due to the TCJA, their reactions to the TCJA in of itself cannot explain our results but comes in addition to the effects which we try to show.

¹¹ Heese et al. (2022) explores the closure of a local newspaper as a shock to the local monitoring by the local media. Our study differs from their approach because we are focusing on the local newspapers' role in disseminating the information that in case of a closure can be more easily taken over by other local newspapers. In additional tests we see that the overall circulation of local newspapers has a stronger effect on the IACs, but also the closures of local newspapers increase the IACs by reducing the circulation.

¹² See Online Appendix B for more details on the regulation and the empirical findings on brokered deposits.

¹³ Because the treatment definitions in Panels A and B of Table 3 capture the DTA levels at different points in time, the sample sizes of banks with DTAs in Q1 2016 do not exactly correspond to the sample size in Panel B.

¹⁴ We calculate the economic significance by multiplying the regression coefficient with the average change of net DTAs due to the TCJA scaled by total assets and relating this with the average change in total deposits: $-2.1868 \times 0.00119 / 0.01455$. See Table 2 for the means of the change in net DTAs and change in total deposits.

¹⁵ Additional tests show common trends in the deposit rates and the change in lending for banks with and without DTAs before the TCJA.

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